



模式识别与计算机视觉

Chapter 10: Alignment and Transformation

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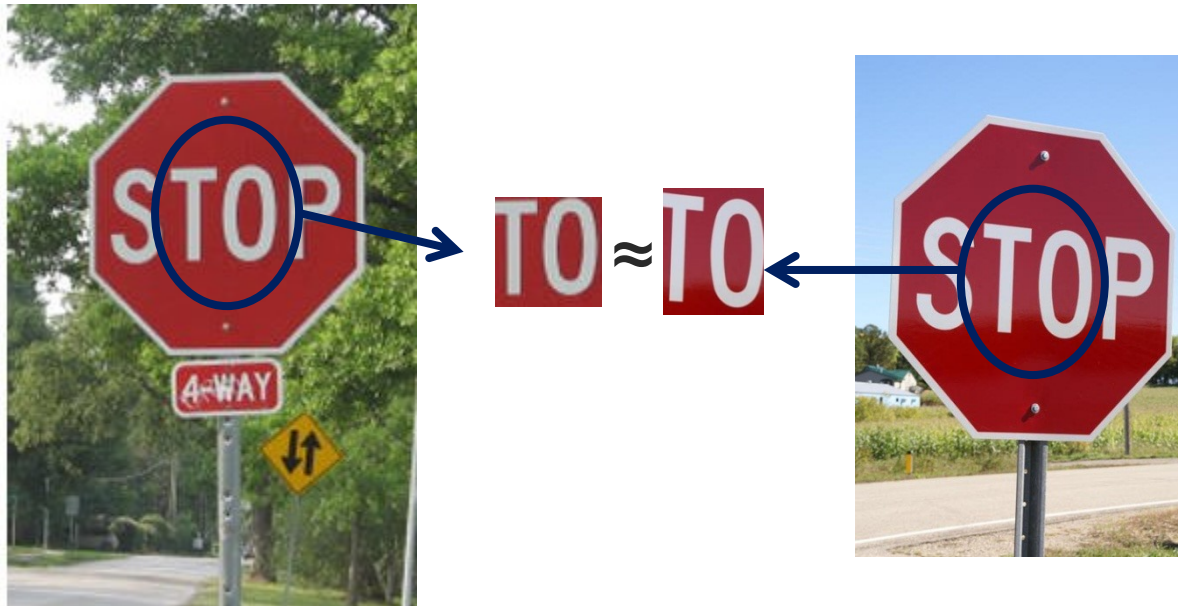
<http://mcg.nju.edu.cn/>



Correspondence and alignment

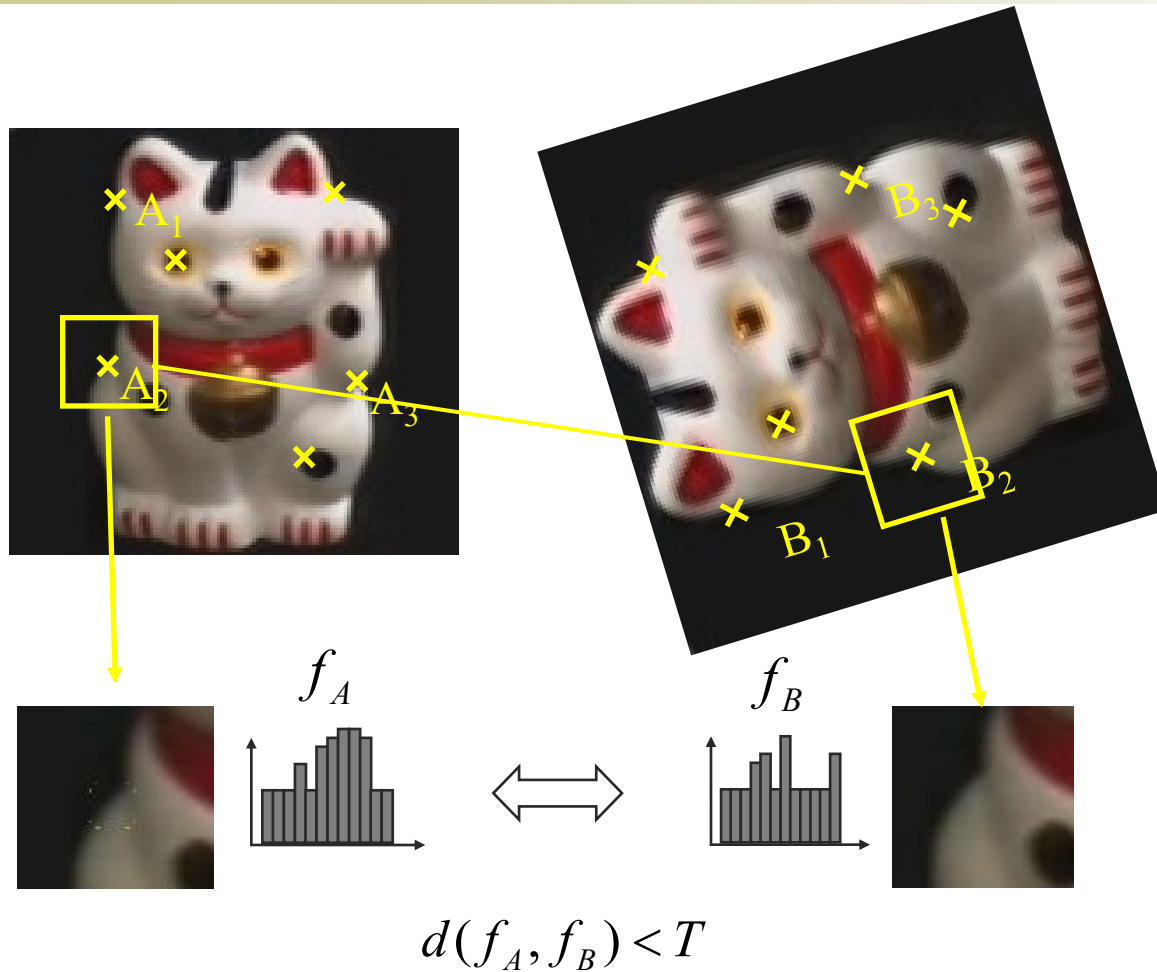


Correspondence: matching points, patches, edges, or regions across images





Recap: Keypoint Matching

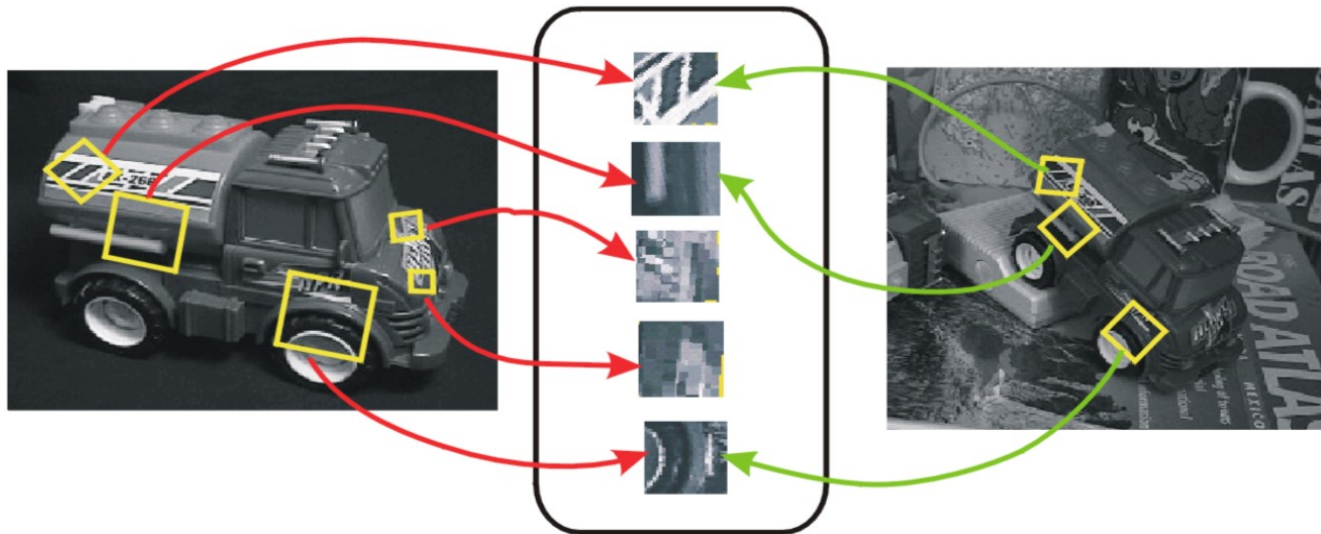


1. Find a set of distinctive keypoints
2. Define a region around each keypoint
3. Extract and normalize the region content
4. Compute a local descriptor from the normalized region
5. Match local descriptors

Invariance vs. Covariance



- **Invariance:**
 - $\text{features}(\text{transform}(\text{image})) = \text{features}(\text{image})$
- **Covariance:**
 - $\text{features}(\text{transform}(\text{image})) = \text{transform}(\text{features}(\text{image}))$



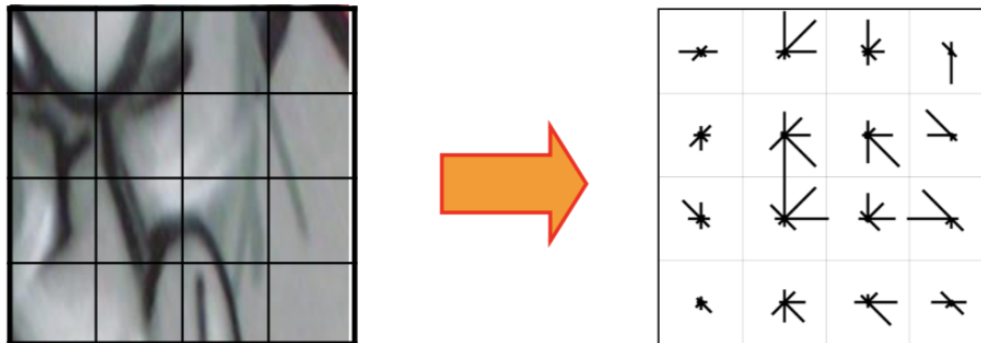
Covariant detection $\Rightarrow$ invariant description



Feature Descriptor: SIFT



- **S**cale **I**nvariant **F**eature **T**ransform
- **Descriptor computation:**
 - Divide patch into 4x4 sub-patches: 16 cells
 - Compute histogram of gradient orientations (8 reference angles) for all pixels inside each sub-patch
 - Resulting descriptor: $4 \times 4 \times 8 = 128$ dimensions



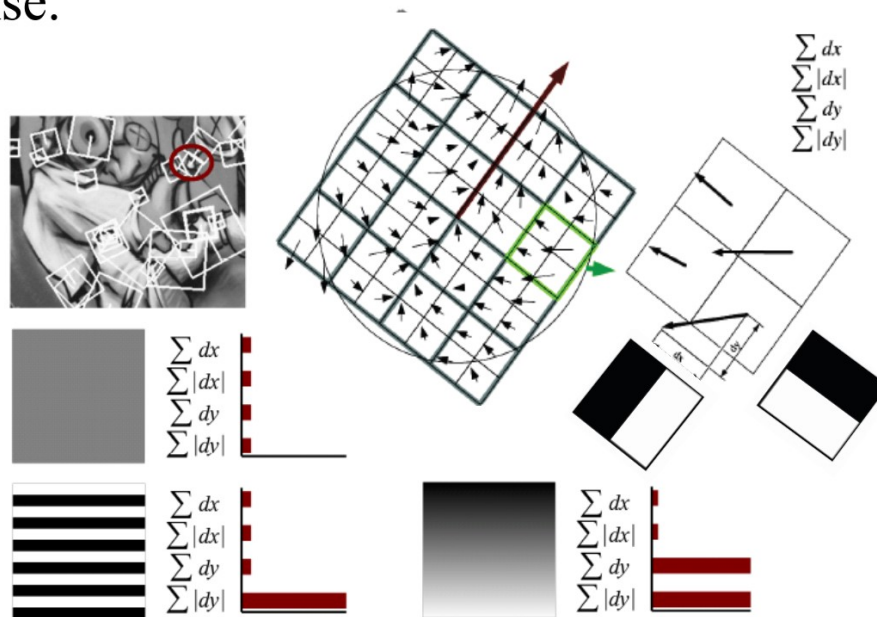
David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#)
IJCV 60 (2), pp. 91-110, 2004.



SURF: Descriptor Extraction



1. Split the interest region (20s x 20s) into 4 x 4 square sub-regions.
2. Calculate Haar wavelet responses dx and dy , and weight the responses with a Gaussian kernel.
3. Sum the response over each sub-region for dx and dy , then sum the absolute value of response.
4. Concatenate summation results in all sub-regions, forming a 64D SURF descriptor.





Today's class



- Introduction to alignment
- Alignment methods
 - Global methods
 - Hypothesize and test
- Image Transformation
 - Common image transformations
 - Examples of solving image alignment
- Homework: Mosaics



Alignment



Alignment: solving the transformation that makes two things match better



T

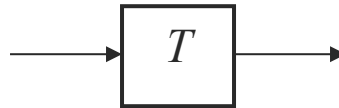




Parametric (global) warping



$$\mathbf{p} = (x, y)$$



$$\mathbf{p}' = (x', y')$$

Transformation T is a coordinate-changing machine:

$$\mathbf{p}' = T(\mathbf{p})$$

What does it mean that T is global?

- Is the same for any point $\mathbf{p}$
- can be described by just a few numbers (parameters)

For linear transformations, we can represent T as a matrix

$$\mathbf{p}' = \mathbf{T}\mathbf{p} \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix}$$



Fitting and Alignment



Fitting: find the parameters of a model that best fit the data.

Alignment: find the parameters of the transformation that best align matched points.



Fitting and Alignment



- Design challenges
 - Design a suitable **goodness of fit** measure
 - Similarity should reflect application goals
 - Encode robustness to outliers and noise
 - Design an **optimization** method
 - Avoid local optima
 - Find best parameters quickly



Fitting and Alignment: Methods



- Global optimization / Search for parameters
 - Least squares fit
 - Robust least squares
 - Iterative closest point (ICP)

- Hypothesize and test
 - Generalized Hough transform
 - RANSAC



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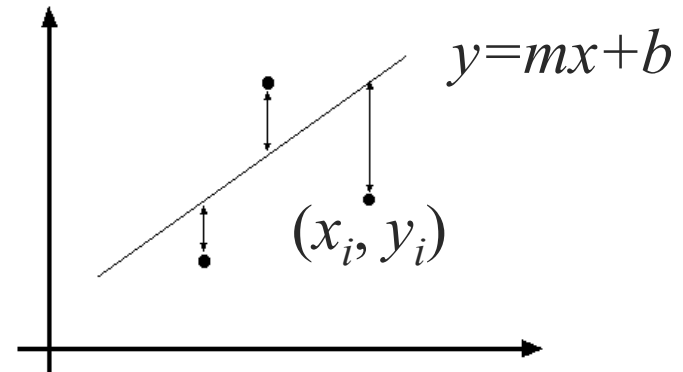


Least squares line fitting



- Data: $(x_1, y_1), \dots, (x_n, y_n)$
- Line equation: $y_i = mx_i + b$
- Find (m, b) to minimize

$$E = \sum_{i=1}^n (y_i - mx_i - b)^2$$



$$E = \sum_{i=1}^n \left(\begin{bmatrix} x_i & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} - y_i \right)^2 = \left\| \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} - \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \right\|^2 = \|\mathbf{A}\mathbf{p} - \mathbf{y}\|^2$$

$$= \mathbf{y}^T \mathbf{y} - 2(\mathbf{A}\mathbf{p})^T \mathbf{y} + (\mathbf{A}\mathbf{p})^T (\mathbf{A}\mathbf{p})$$

$$\frac{dE}{dp} = 2\mathbf{A}^T \mathbf{A}\mathbf{p} - 2\mathbf{A}^T \mathbf{y} = 0$$

$$\mathbf{A}^T \mathbf{A}\mathbf{p} = \mathbf{A}^T \mathbf{y} \Rightarrow \mathbf{p} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$

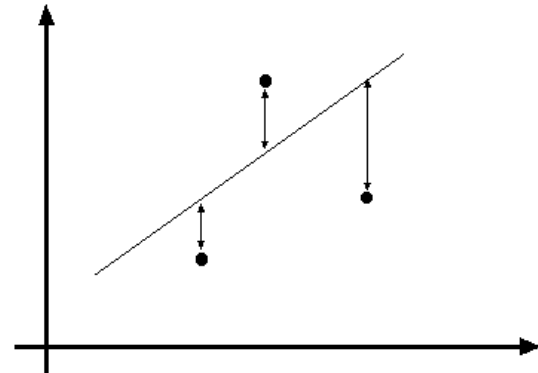
$$\text{Matlab: } \mathbf{p} = \mathbf{A} \setminus \mathbf{y};$$



Problem with “vertical” least squares



- Not rotation-invariant
- Fails completely for vertical lines



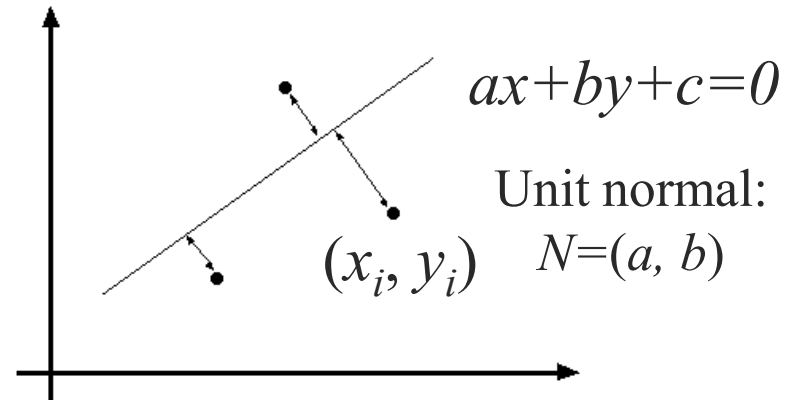


Total least squares



If $(a^2+b^2=1)$ then

Distance between point (x_i, y_i) and line $ax+by+c=0$ is $|ax_i + by_i + c|$



proof:

<http://mathworld.wolfram.com/Point-LineDistance2-Dimensional.html>



Total least squares

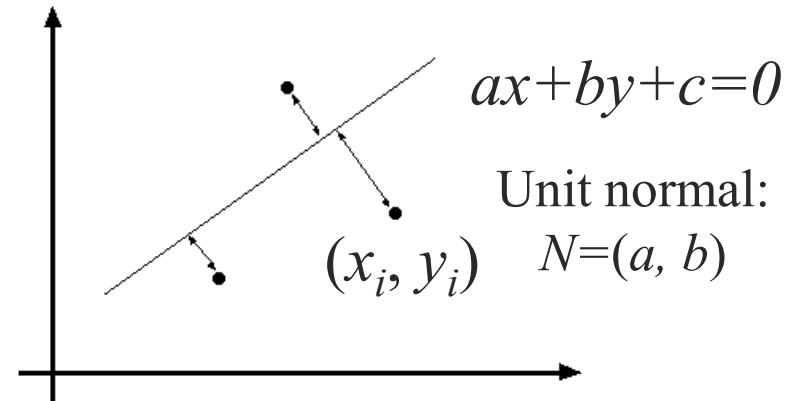


If $(a^2+b^2=1)$ then

Distance between point (x_i, y_i) and line $ax+by+c=0$ is $|ax_i + by_i + c|$

Find (a, b, c) to minimize the sum of squared perpendicular distances

$$E = \sum_{i=1}^n (ax_i + by_i + c)^2$$





Total least squares



Find (a, b, c) to minimize the sum of squared perpendicular distances

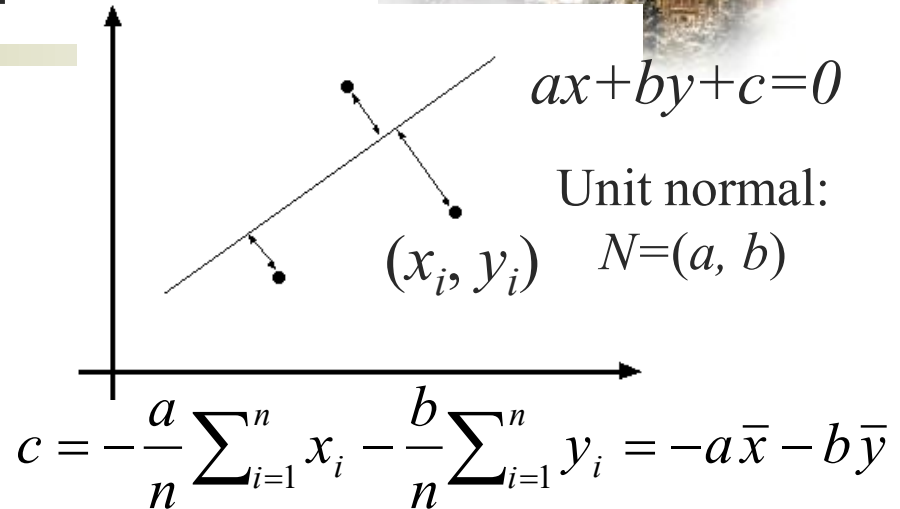
$$E = \sum_{i=1}^n (ax_i + by_i + c)^2$$

$$\frac{\partial E}{\partial c} = \sum_{i=1}^n 2(ax_i + by_i + c) = 0$$

$$E = \sum_{i=1}^n (a(x_i - \bar{x}) + b(y_i - \bar{y}))^2 = \left\| \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \right\|^2 = \mathbf{p}^T \mathbf{A}^T \mathbf{A} \mathbf{p}$$

$$\text{minimize } \mathbf{p}^T \mathbf{A}^T \mathbf{A} \mathbf{p} \quad \text{s.t. } \mathbf{p}^T \mathbf{p} = 1 \quad \Rightarrow \quad \text{minimize } \frac{\mathbf{p}^T \mathbf{A}^T \mathbf{A} \mathbf{p}}{\mathbf{p}^T \mathbf{p}}$$

Solution is eigenvector corresponding to smallest eigenvalue of $\mathbf{A}^T \mathbf{A}$





Recap: Two Common Optimization Problems



Problem statement

$$\text{minimize } \|\mathbf{Ax} - \mathbf{b}\|^2$$

least squares solution to $\mathbf{Ax} = \mathbf{b}$

Solution

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

$$\mathbf{x} = \mathbf{A} \setminus \mathbf{b} \quad (\text{matlab})$$

Problem statement

$$\text{minimize } \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} \quad \text{s.t. } \mathbf{x}^T \mathbf{x} = 1$$

$$\text{minimize } \frac{\mathbf{x}^T \mathbf{A}^T \mathbf{Ax}}{\mathbf{x}^T \mathbf{x}}$$

non - trivial lsq solution to $\mathbf{Ax} = 0$

Solution

$$[\mathbf{v}, \lambda] = \text{eig}(\mathbf{A}^T \mathbf{A})$$

$$\lambda_1 < \lambda_{2..n} : \mathbf{x} = \mathbf{v}_1$$



Least squares (global) optimization



Good

- Clearly specified objective
- Optimization is easy

Bad

- May not be what you want to optimize
- Sensitive to outliers
 - Bad matches, extra points
- Doesn't allow you to get multiple good fits
 - Detecting multiple objects, lines, etc.



Other ways to search for parameters



■ Line search

1. For each parameter, step through values and choose value that gives best fit
2. Repeat (1) until no parameter changes

■ Grid search

1. Propose several sets of parameters, evenly sampled in the joint set
2. Choose best (or top few) and sample joint parameters around the current best; repeat

■ Gradient descent

1. Provide initial position (e.g., random)
2. Locally search for better parameters by following gradient



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 - **Hypothesize and test**
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Hypothesize and test



1. Propose parameters

- Try all possible
- Each point votes for all consistent parameters
- Repeatedly sample enough points to solve for parameters

2. Score the given parameters

- Number of consistent points, possibly weighted by distance

3. Choose from among the set of parameters

- Global or local maximum of scores

4. Possibly refine parameters using inliers



Hough Transform: Outline



1. Create a grid of parameter values
2. Each point votes for a set of parameters, incrementing those values in grid
3. Find maximum or local maxima in grid



Hough Space

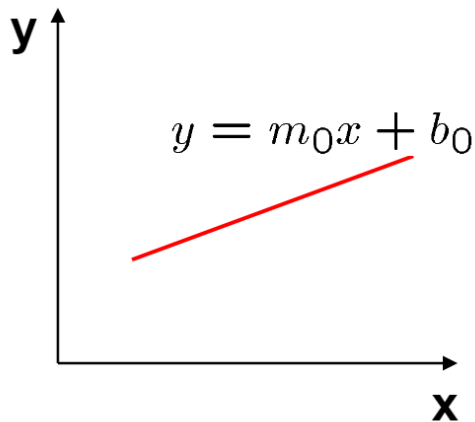
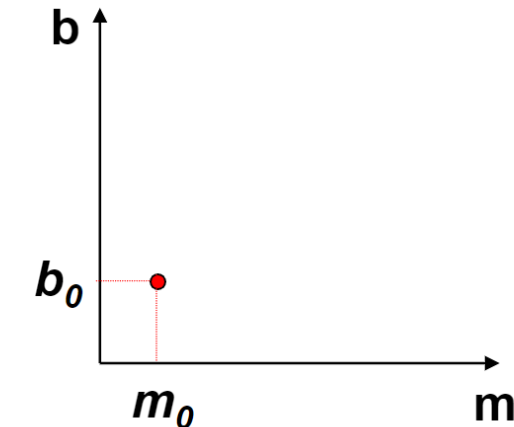


Image space

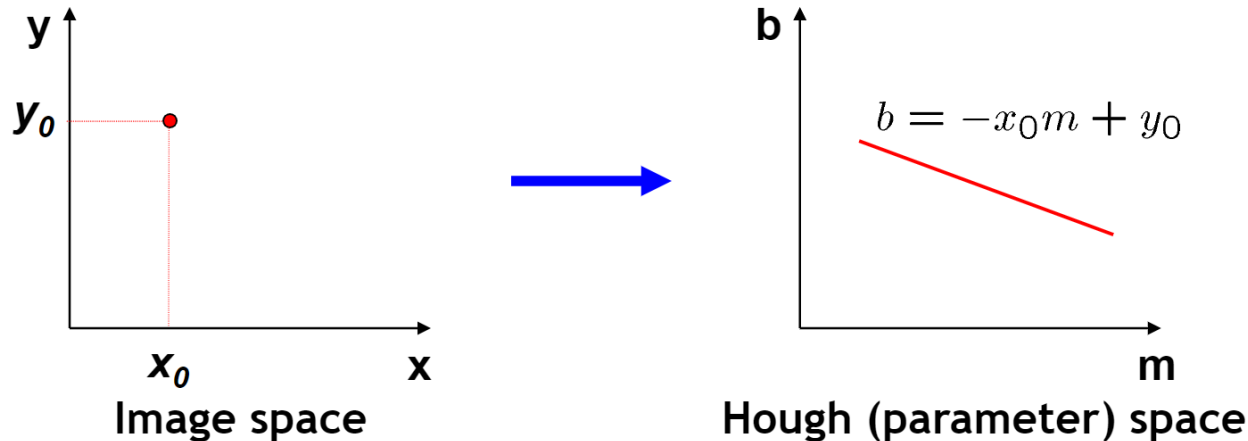


Hough (parameter) space

- **Connection between image (x,y) and Hough (m,b) spaces**
 - A line in the image corresponds to a point in Hough space.
 - To go from image space to Hough space:
 - Given a set of points (x,y) , find all (m,b) such that $y = mx + b$



Hough Space



- **Connection between image (x,y) and Hough (m,b) spaces**
 - A line in the image corresponds to a point in Hough space.
 - To go from image space to Hough space:
 - Given a set of points (x,y) , find all (m,b) such that $y = mx + b$
 - What does a point (x_0, y_0) in the image space map to?
 - Answer: the solutions of $b = -x_0m + y_0$
 - This is a line in Hough space.

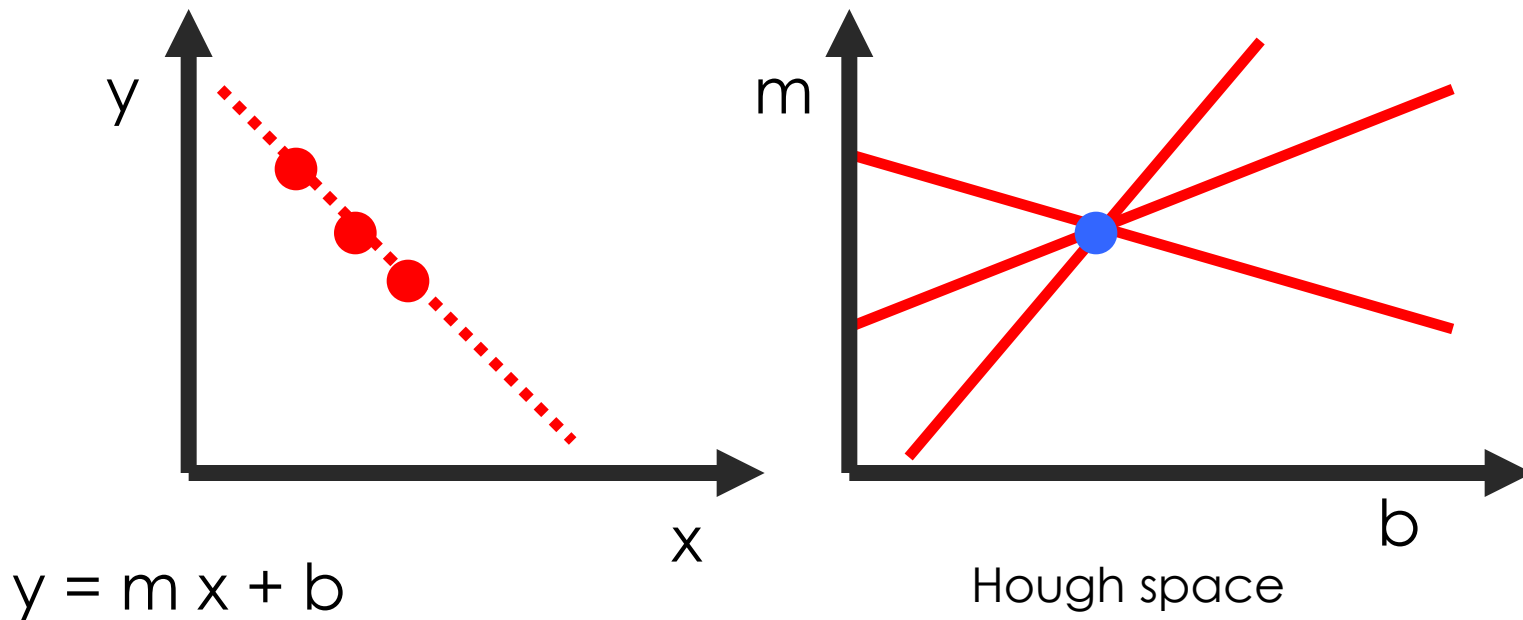


Hough transform



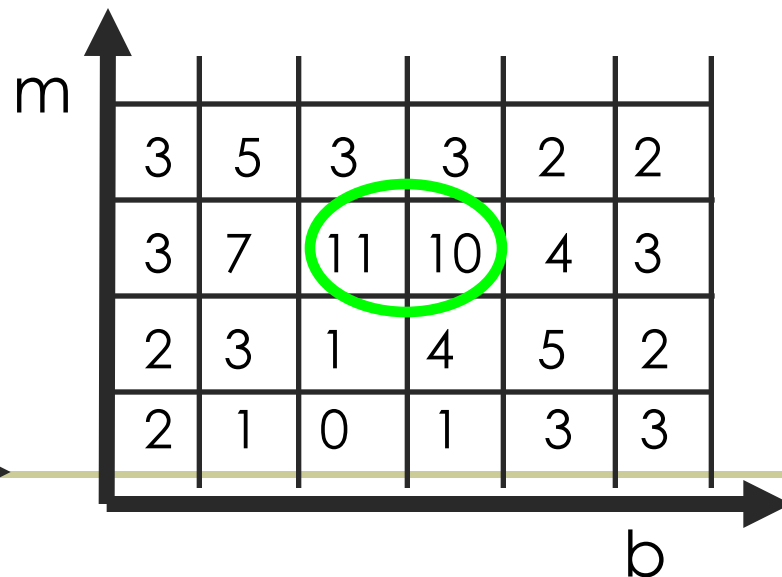
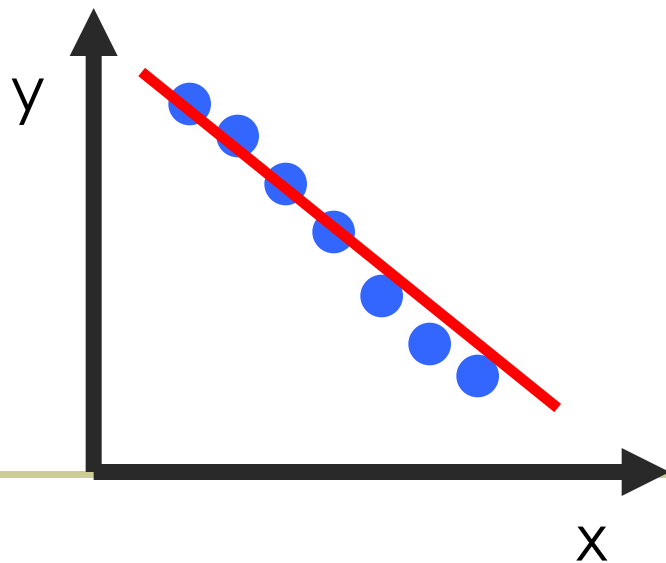
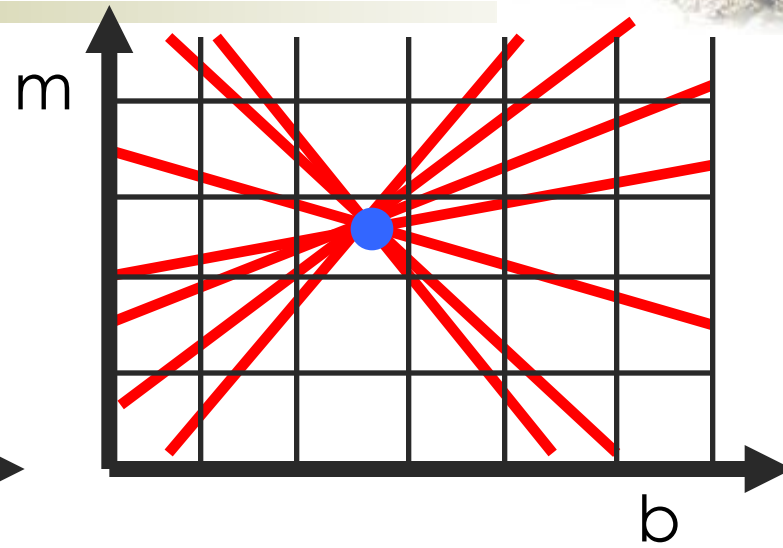
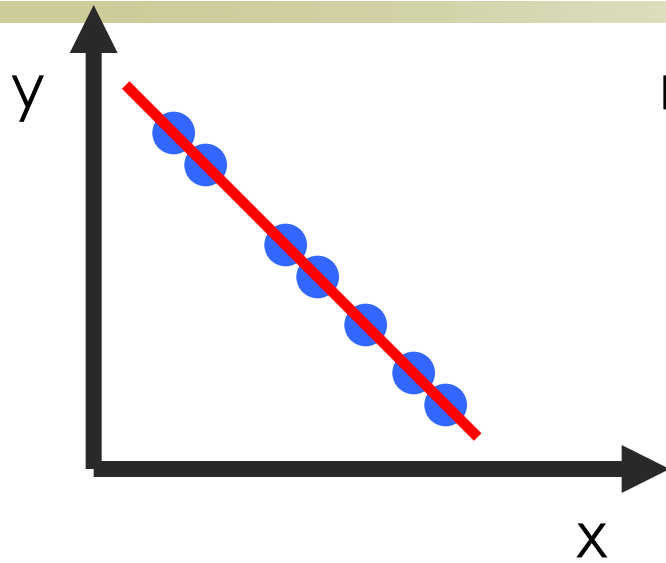
P.V.C. Hough, *Machine Analysis of Bubble Chamber Pictures*, Proc. Int. Conf. High Energy Accelerators and Instrumentation, 1959

Given a set of points, find the curve or line that explains the data points best





Hough transform

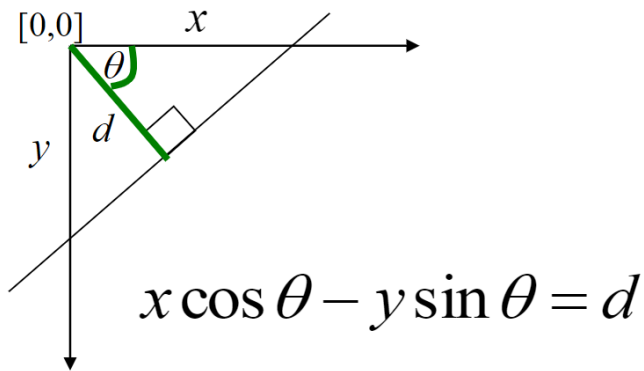




Polar Representation for Lines



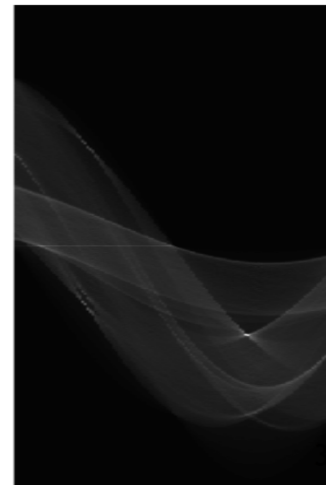
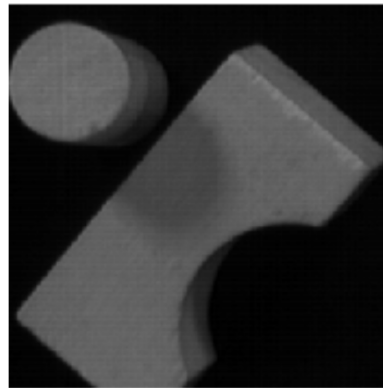
- Issues with usual (m,b) parameter space: can take on infinite values, undefined for vertical lines.



d : perpendicular distance from line to origin

θ : angle the perpendicular makes with the x-axis

- Point in image space
 $\Rightarrow$ Sinusoid segment in Hough space





Hough Transform Algorithm



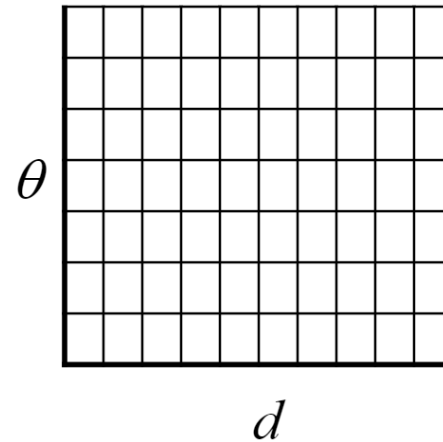
Using the polar parameterization:

$$x \cos \theta + y \sin \theta = d$$

Basic Hough transform algorithm

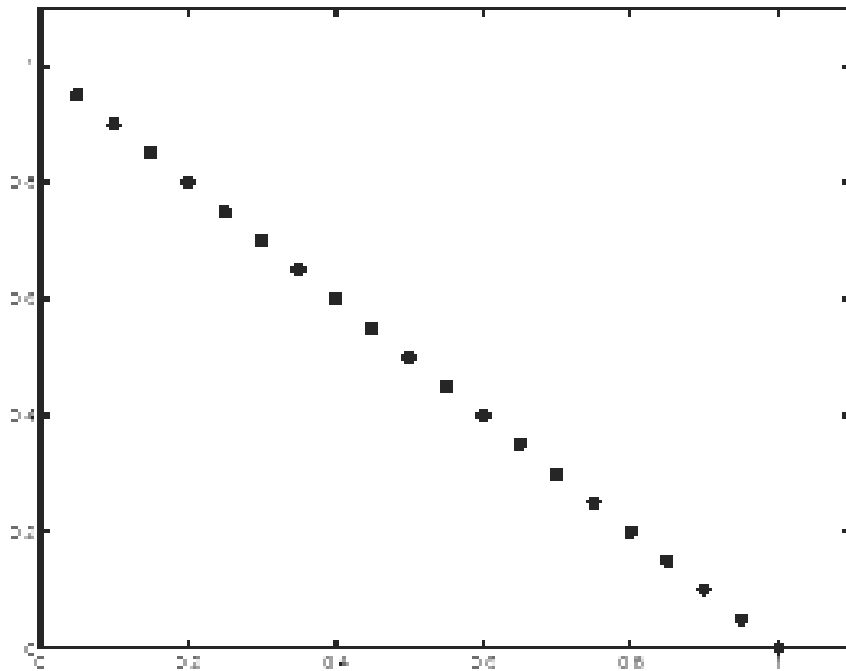
1. Initialize $H[d, \theta] = 0$.
 2. For each edge point (x, y) in the image
for $\theta = 0$ to 180 // some quantization
 $d = x \cos \theta + y \sin \theta$
 $H[d, \theta] += 1$
 3. Find the value(s) of (d, θ) where $H[d, \theta]$ is maximal.
 4. The detected line in the image is given by $d = x \cos \theta + y \sin \theta$
- Time complexity (in terms of number of votes)?

H : accumulator array (votes)

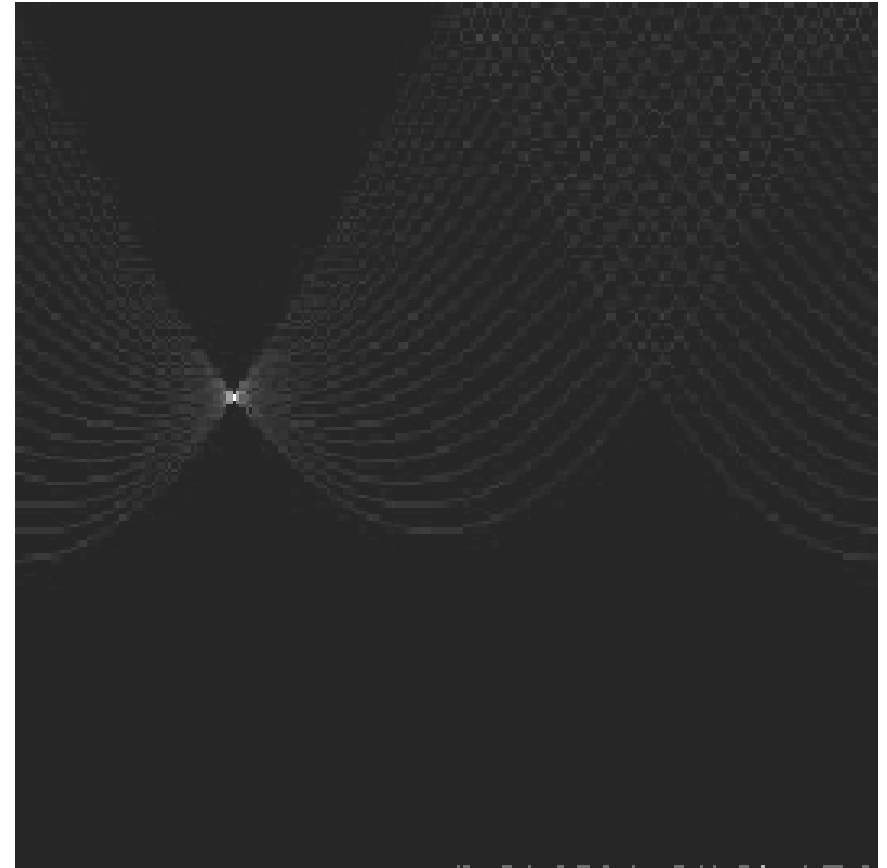




Hough transform - experiments



features



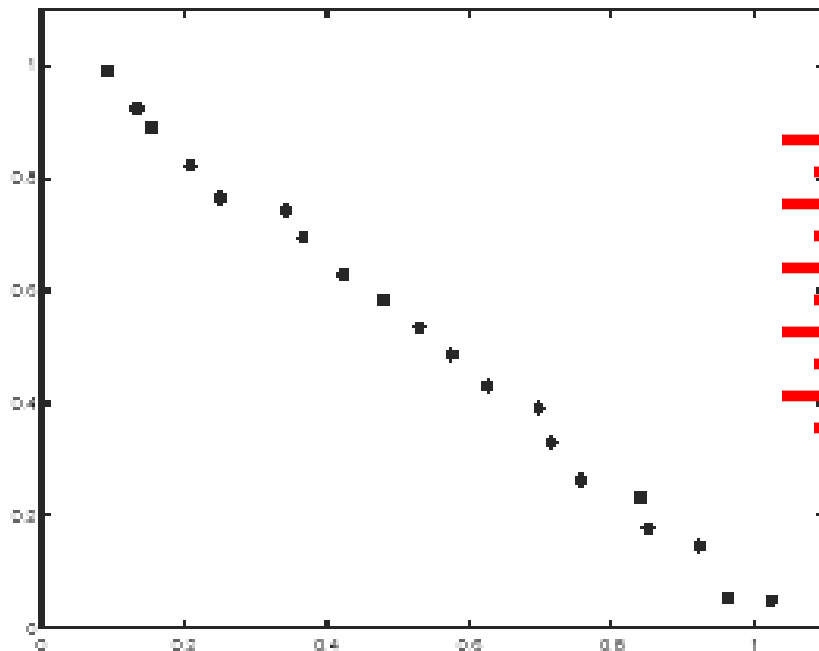
votes



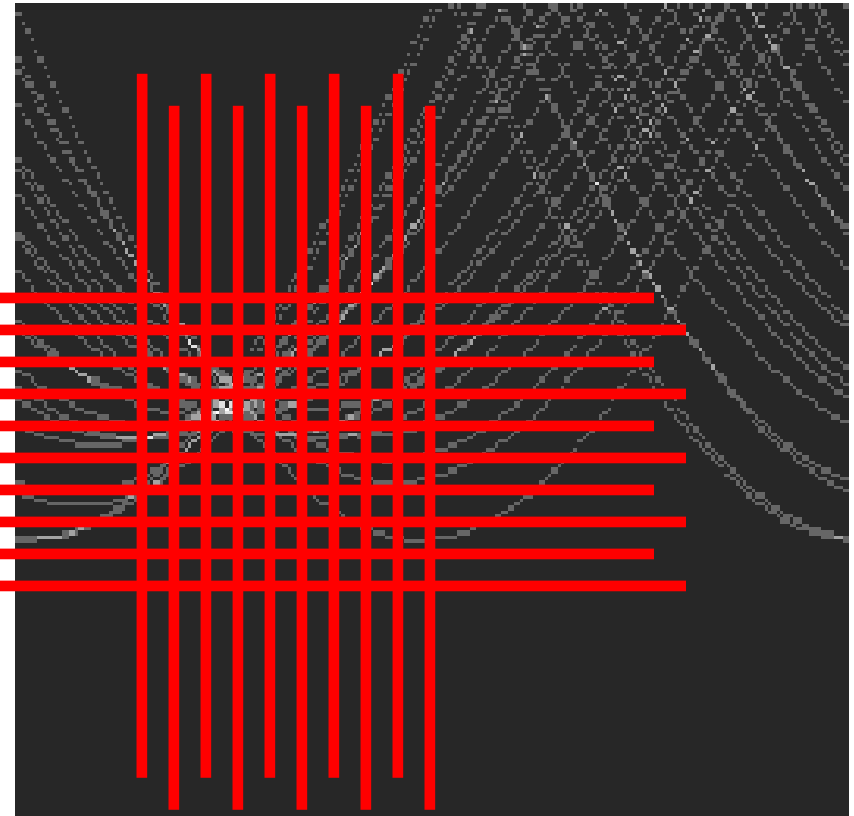
Hough transform - experiments



Noisy data



features

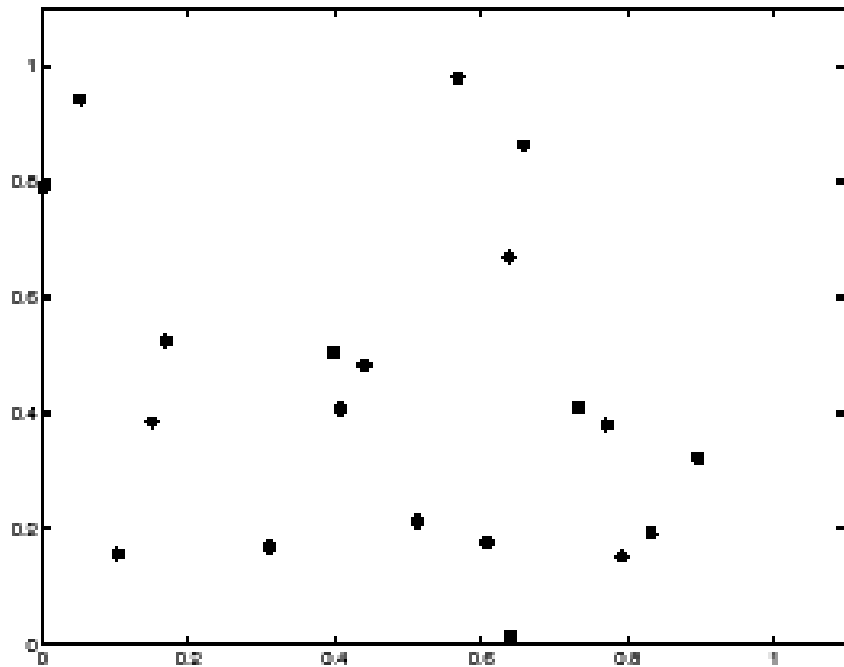


votes

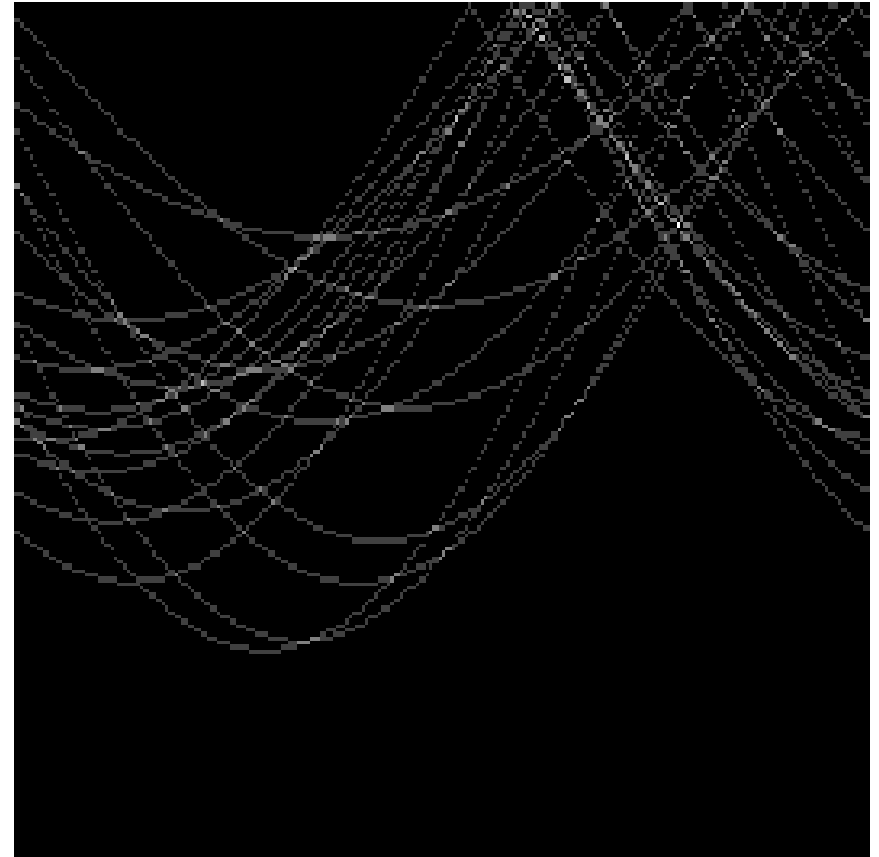
Need to adjust grid size or smooth



Hough transform - experiments



features



votes

Issue: spurious peaks due to uniform noise



1. Image $\rightarrow$ Canny





2. Canny $\rightarrow$ Hough votes

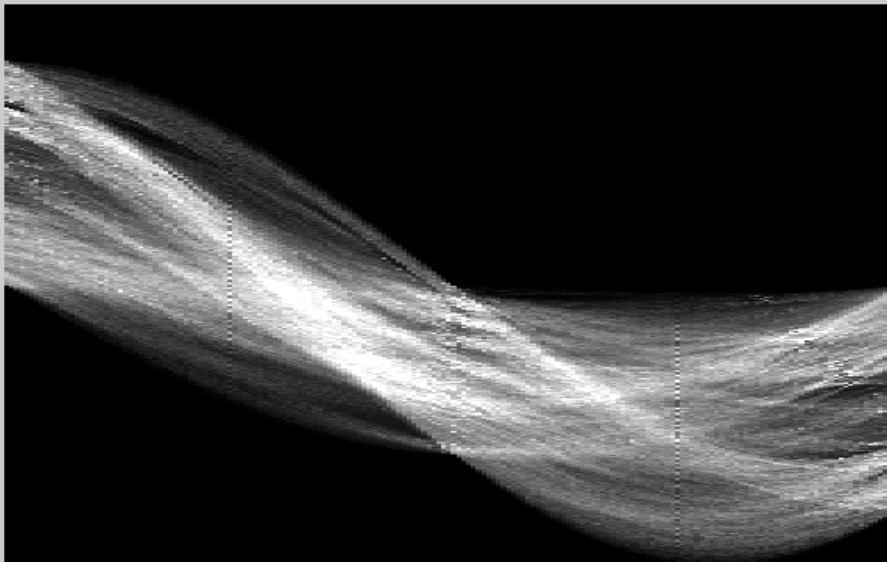




3. Hough votes $\rightarrow$ Edges

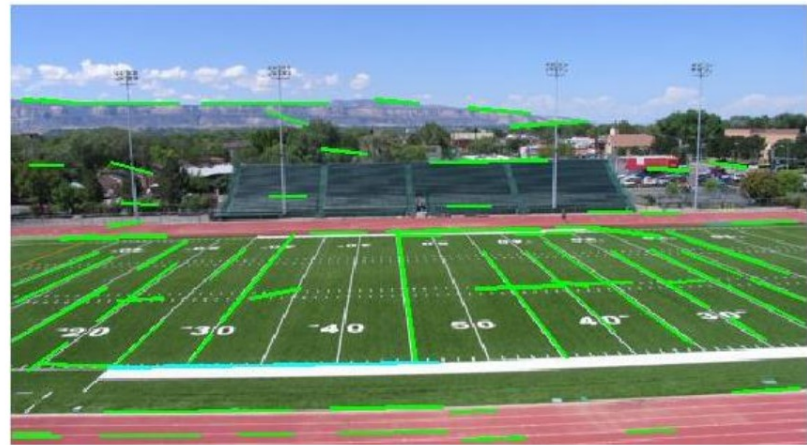
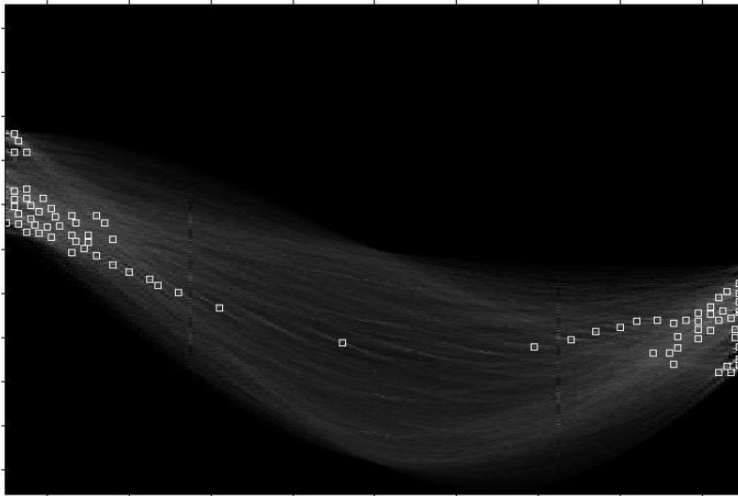
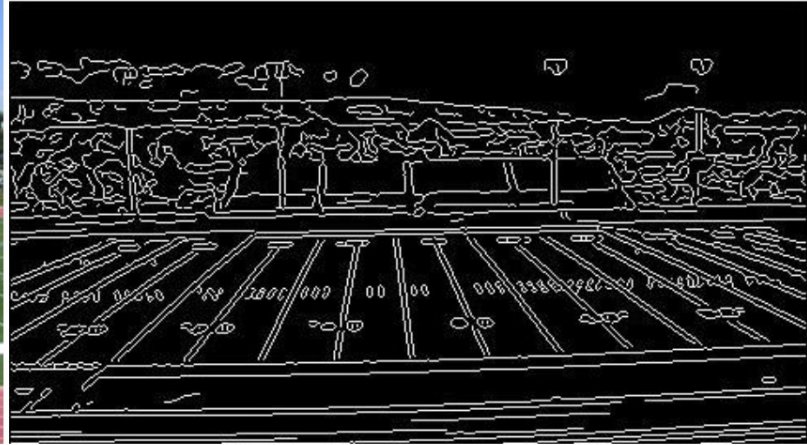


Find peaks and post-process





Real-World Example



Showing longest segments found



Hough transform: pros and cons



Pros

- All points are processed independently, so can cope with occlusion, gaps
- Some robustness to noise: noise points unlikely to contribute *consistently* to any single bin
- Can detect multiple instances of a model in a single pass

Cons

- Complexity of search time increases exponentially with the number of model parameters
- Non-target shapes can produce spurious peaks in parameter space
- Quantization: can be tricky to pick a good grid size



RANSAC



- RANdom Sample Consensus
- **Approach:** we want to avoid the impact of outliers, so let's look for “inliers”, and use those only.
- **Intuition:** if an outlier is chosen to compute the current fit, then the resulting **line** (**transformation**) won't have much support from rest of the **points** (**matches**).



RANSAC



RANSAC loop:

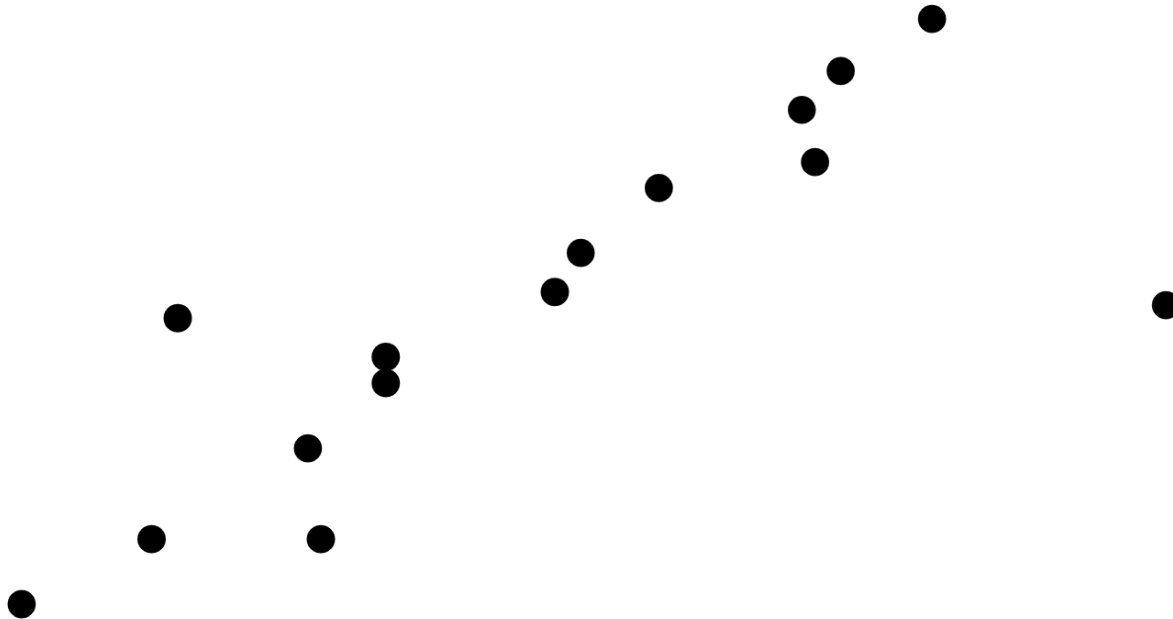
1. Randomly select a *seed group* of points on which to base transformation estimate (e.g., a group of matches)
 2. Compute transformation from seed group
 3. Find *inliers* to this transformation
 4. If the number of inliers is sufficiently large, re-compute least-squares estimate of transformation on all of the inliers
- Keep the transformation with the largest number of inliers



RANSAC Line Fitting Example



- Task: Estimate the best line
 - *How many points do we need to estimate the line?*

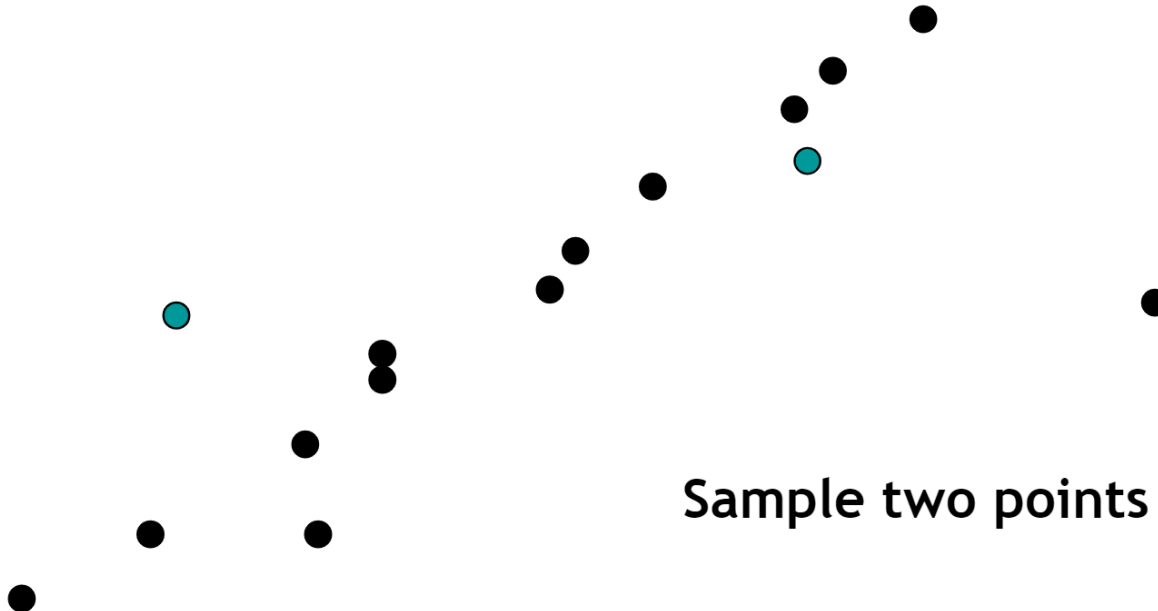




RANSAC Line Fitting Example



- Task: Estimate the best line

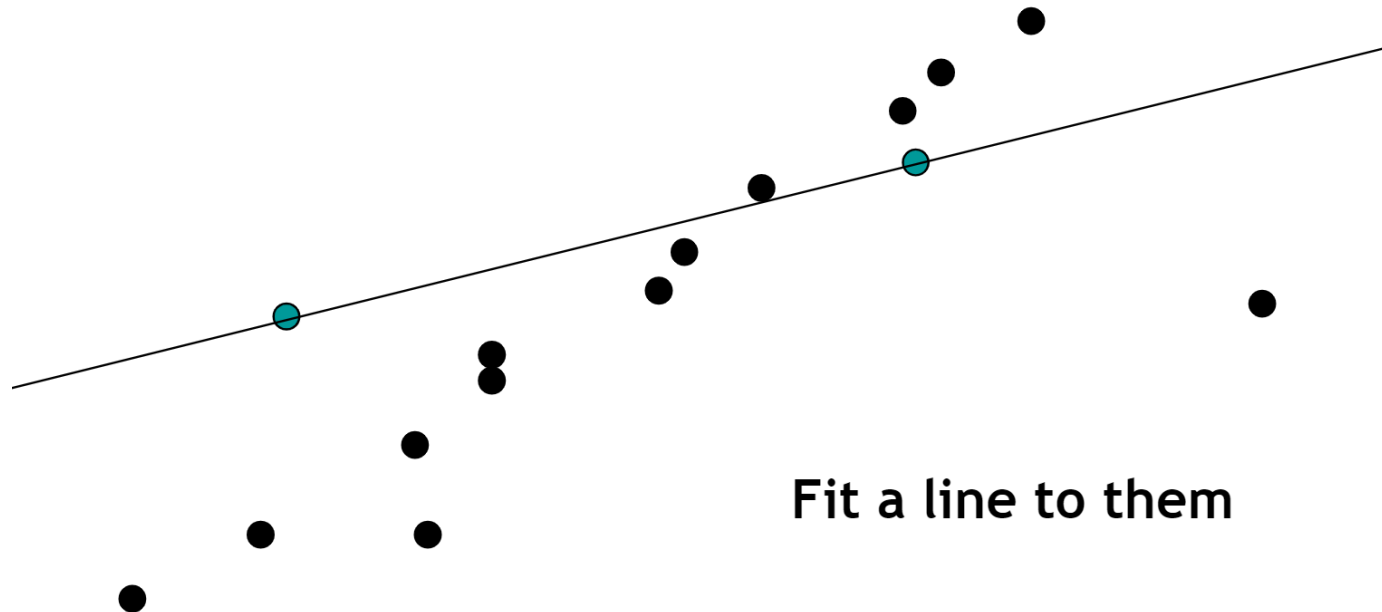




RANSAC Line Fitting Example



- Task: Estimate the best line

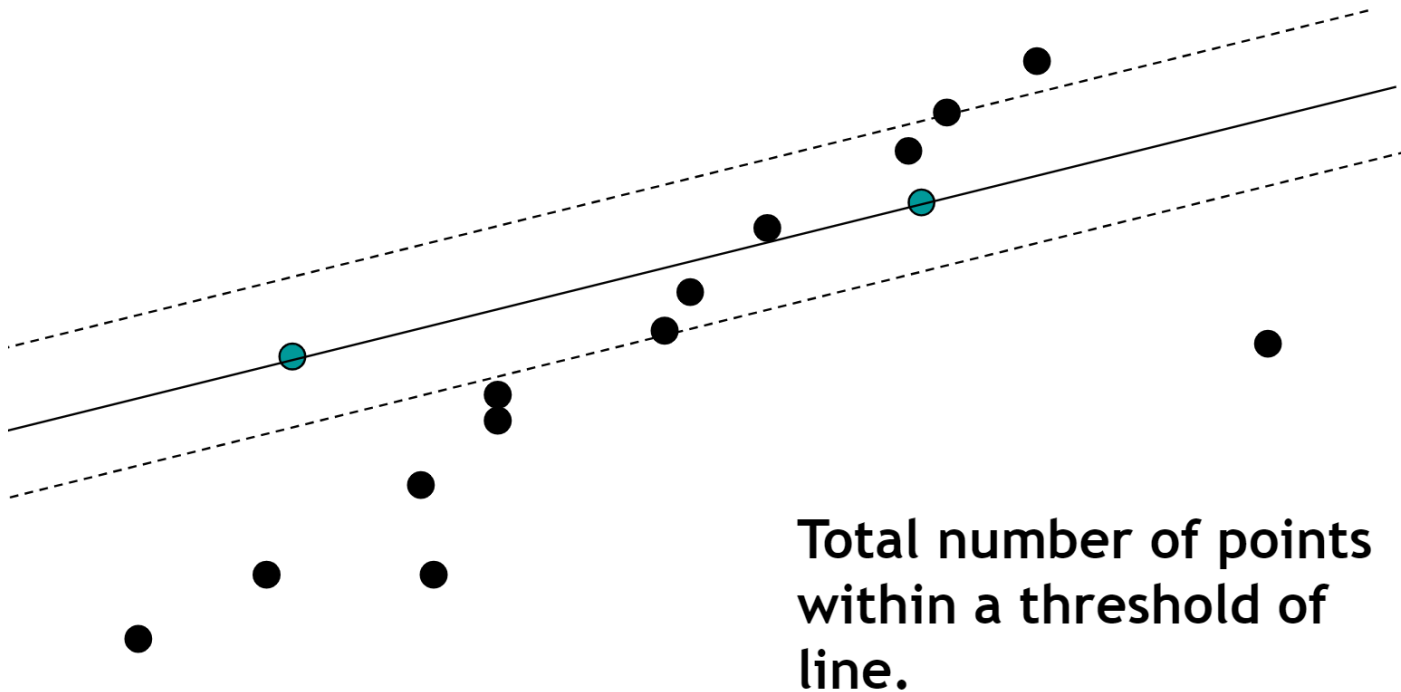




RANSAC Line Fitting Example



- Task: Estimate the best line

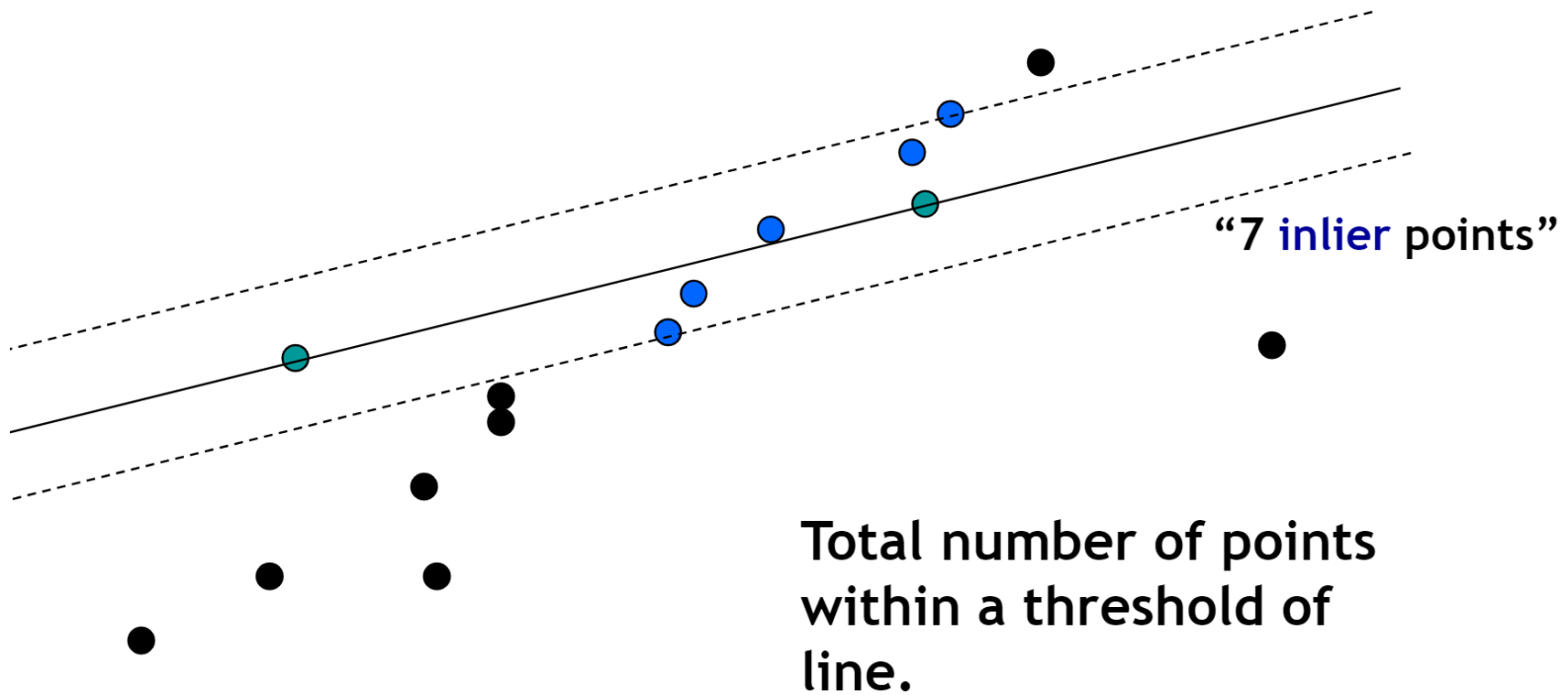




RANSAC Line Fitting Example



- Task: Estimate the best line

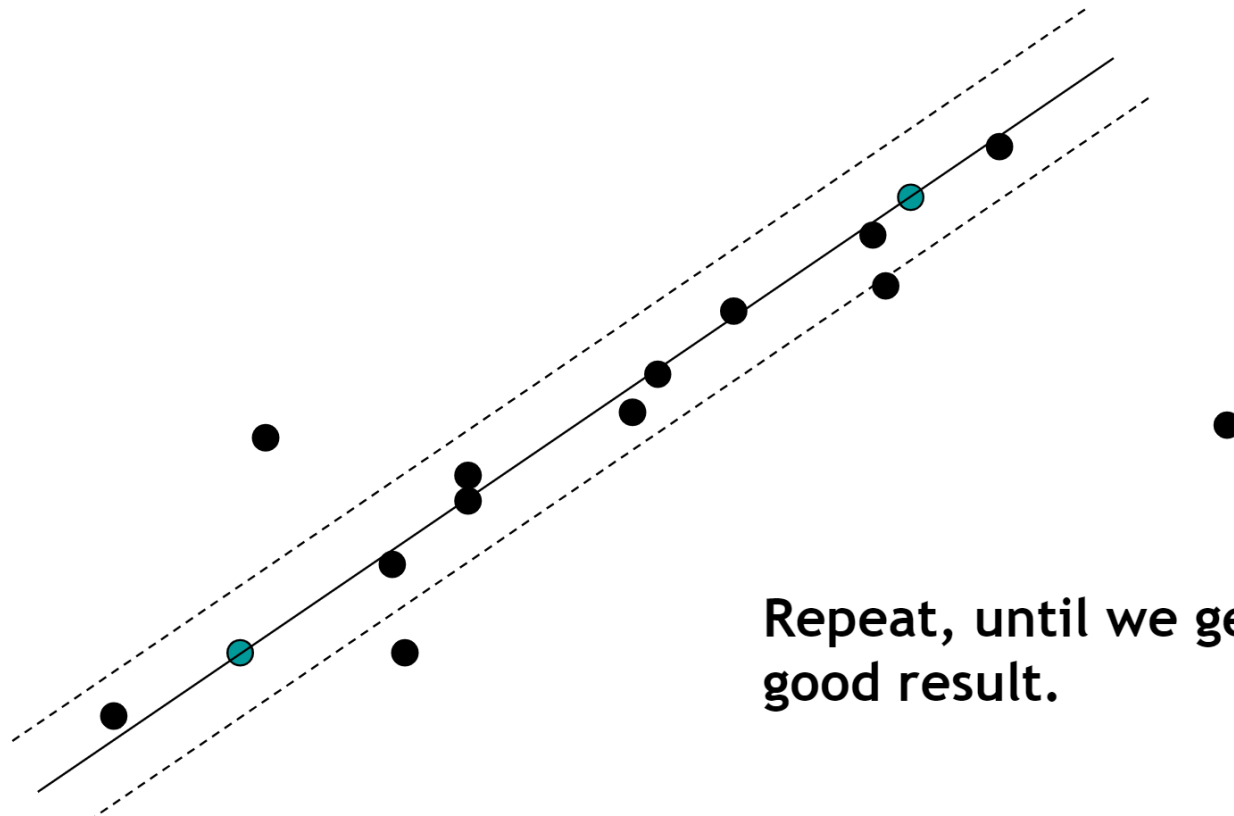




RANSAC Line Fitting Example



- Task: Estimate the best line



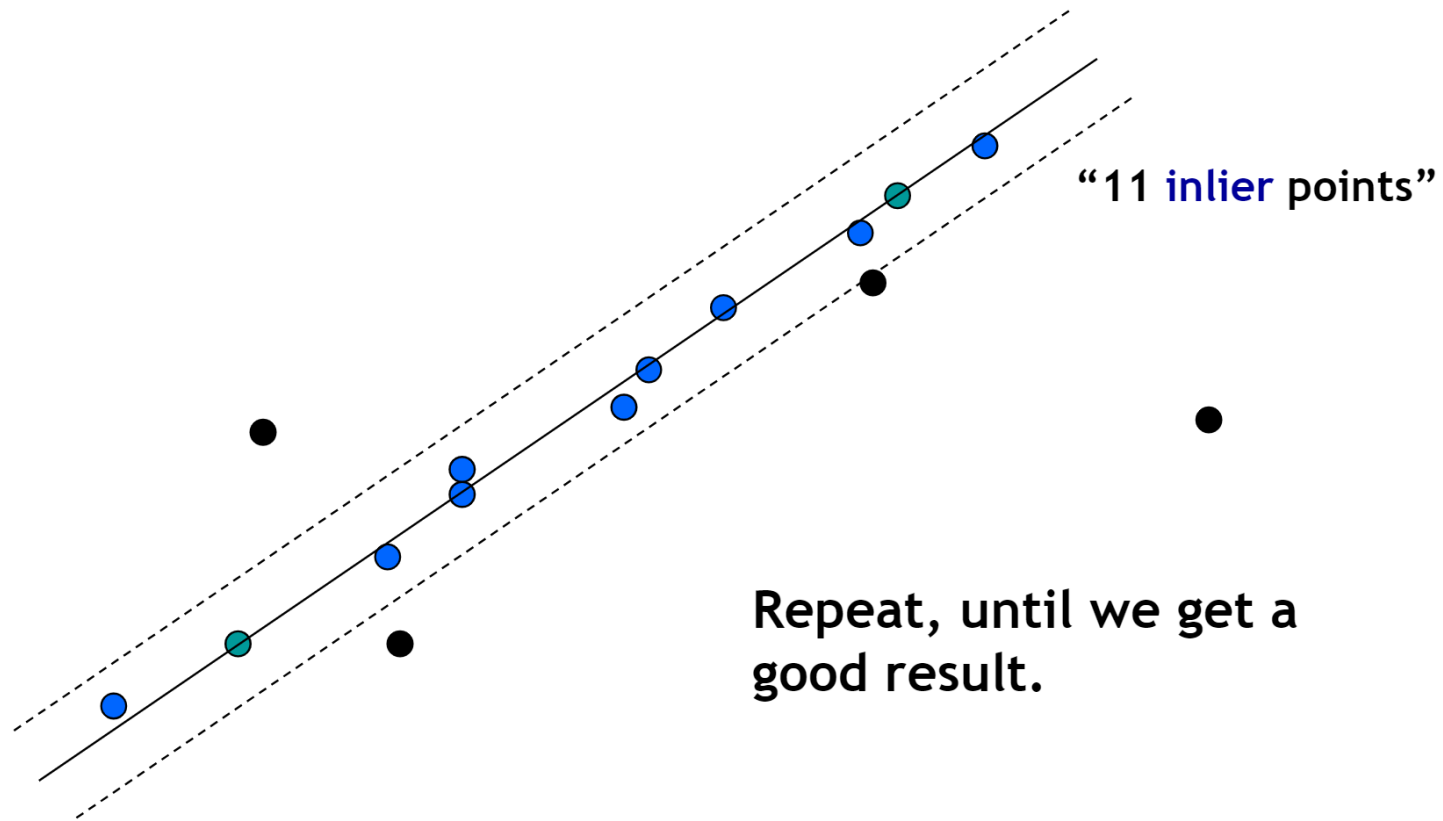
Repeat, until we get a good result.



RANSAC Line Fitting Example



- Task: Estimate the best line





RANSAC: How many samples?



- How many samples are needed?
 - Suppose w is fraction of inliers (points from line).
 - n points needed to define hypothesis (2 for lines)
 - k samples chosen.
 - Prob. that a single sample of n points is correct: w^n
 - Prob. that all k samples fail is: $(1 - w^n)^k$
- ⇒ Choose k high enough to keep this below desired failure rate.



RANSAC: Computed k ($p=0.99$)



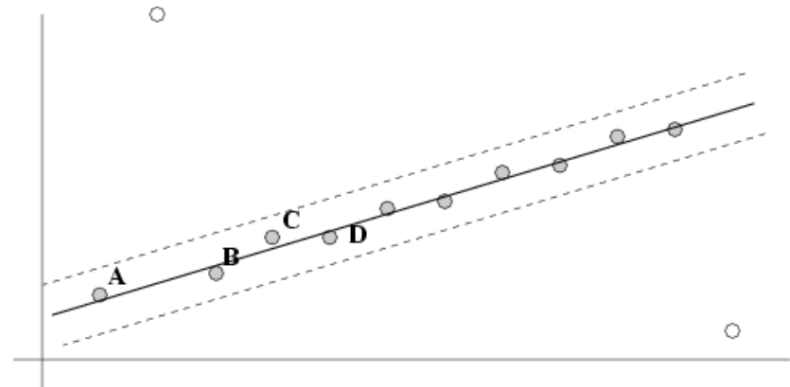
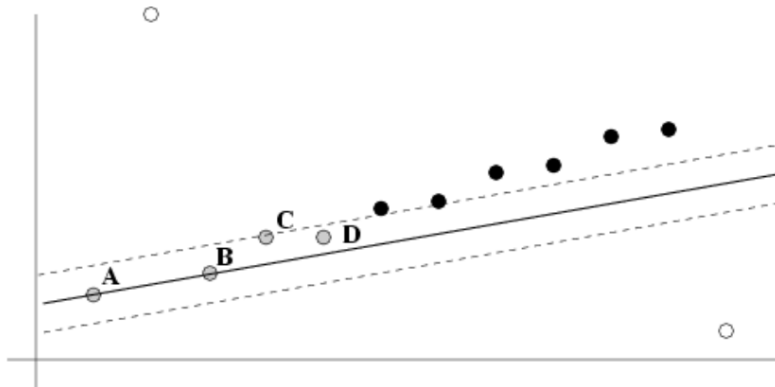
Sample size n	Proportion of outliers						
	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177



After RANSAC



- RANSAC divides data into inliers and outliers and yields estimate computed from minimal set of inliers.
- Improve this initial estimate with estimation over all inliers (e.g. with standard least-squares minimization).
- But this may change inliers, so alternate fitting with reclassification as inlier/outlier.





RANSAC Example

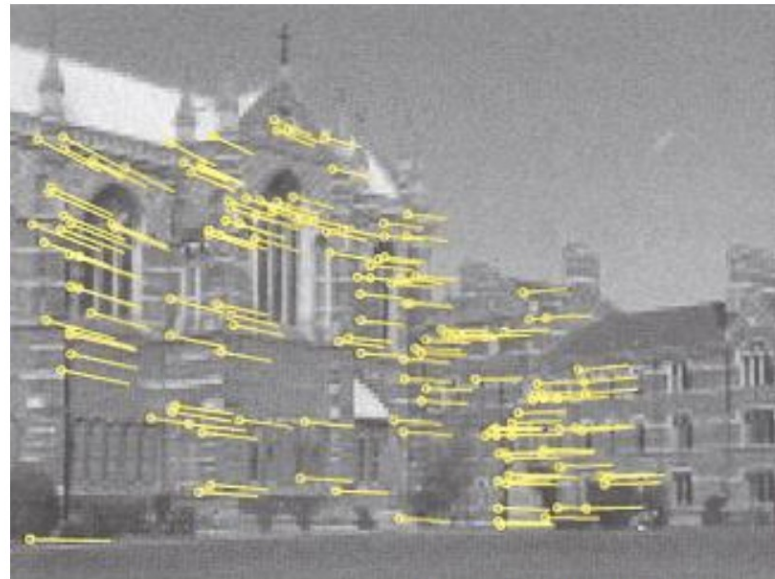


- Find best stereo match within a square search window (here 300 pixels^2)
- Global transformation model: epipolar geometry

before RANSAC



after RANSAC



Images from Hartley & Zisserman



RANSAC conclusions



Good

- Robust to outliers
- Applicable for larger number of objective function parameters than Hough transform
- Optimization parameters are easier to choose than Hough transform

Bad

- Computational time grows quickly with fraction of outliers and number of parameters
- Not as good for getting multiple fits (though one solution is to remove inliers after each fit and repeat)

Common applications

- **Computing a homography (e.g., image stitching)**
- Estimating fundamental matrix (relating two views)



Today's class



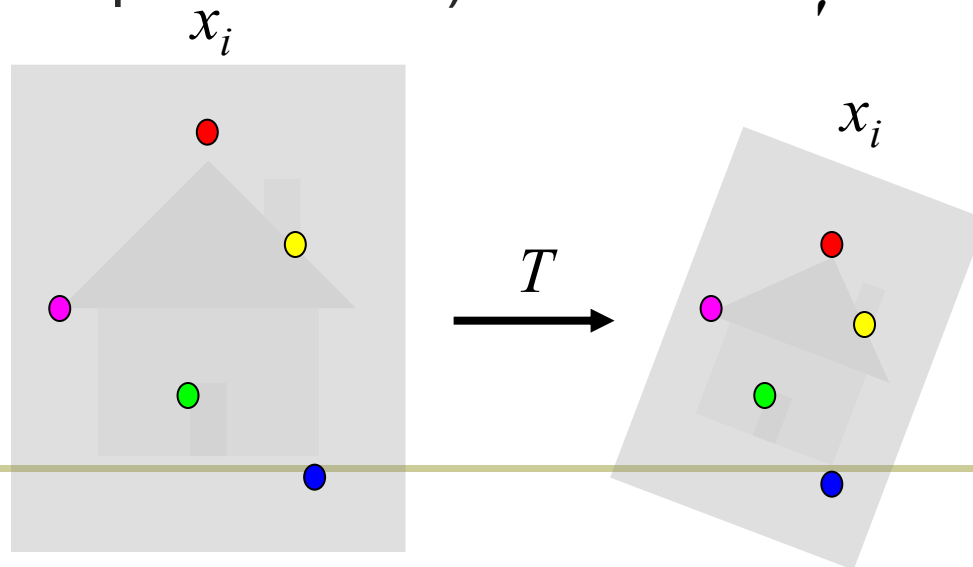
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Alignment problem



- We have previously considered how to **fit a model to image evidence**
 - e.g., a line to edge points
- In alignment, we will fit the parameters of some **transformation** according to a set of matching feature pairs (“correspondences”).





Common transformations



original

Transformed



translation



rotation



aspect



affine



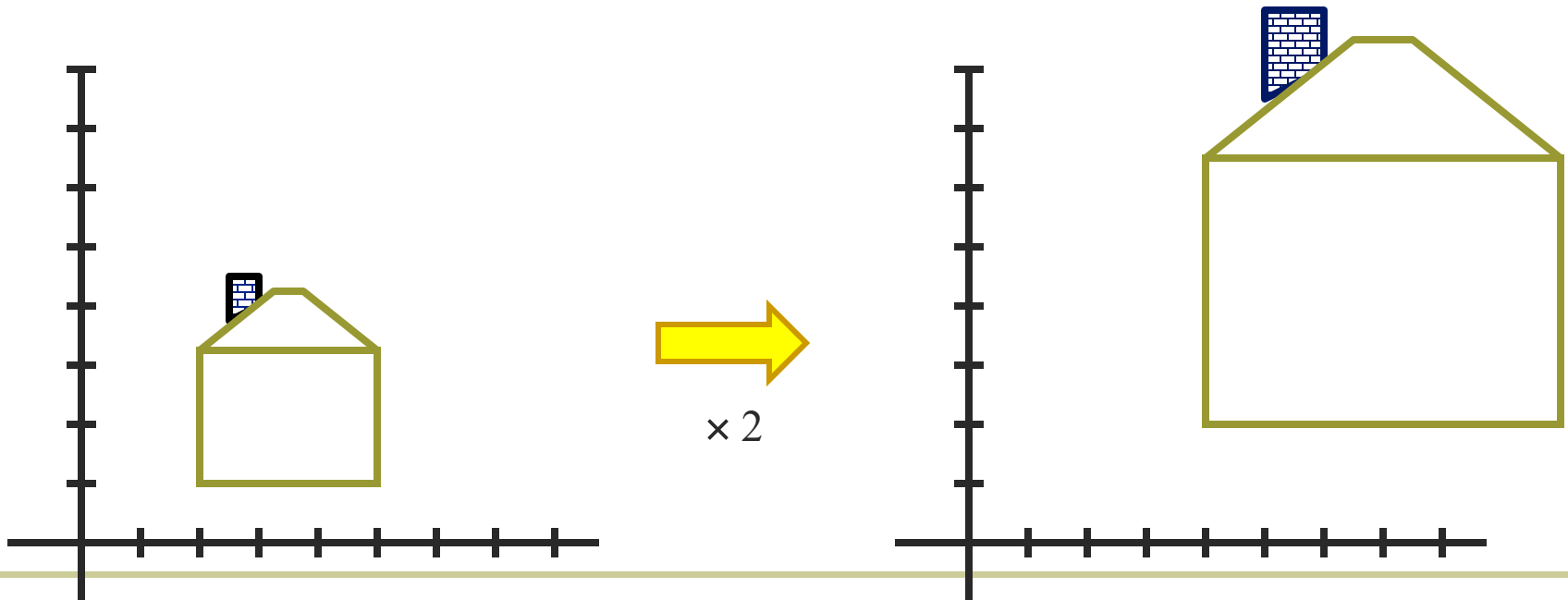
perspective



Scaling



- *Scaling* a coordinate means multiplying each of its components by a scalar
- *Uniform scaling* means this scalar is the same for all components:

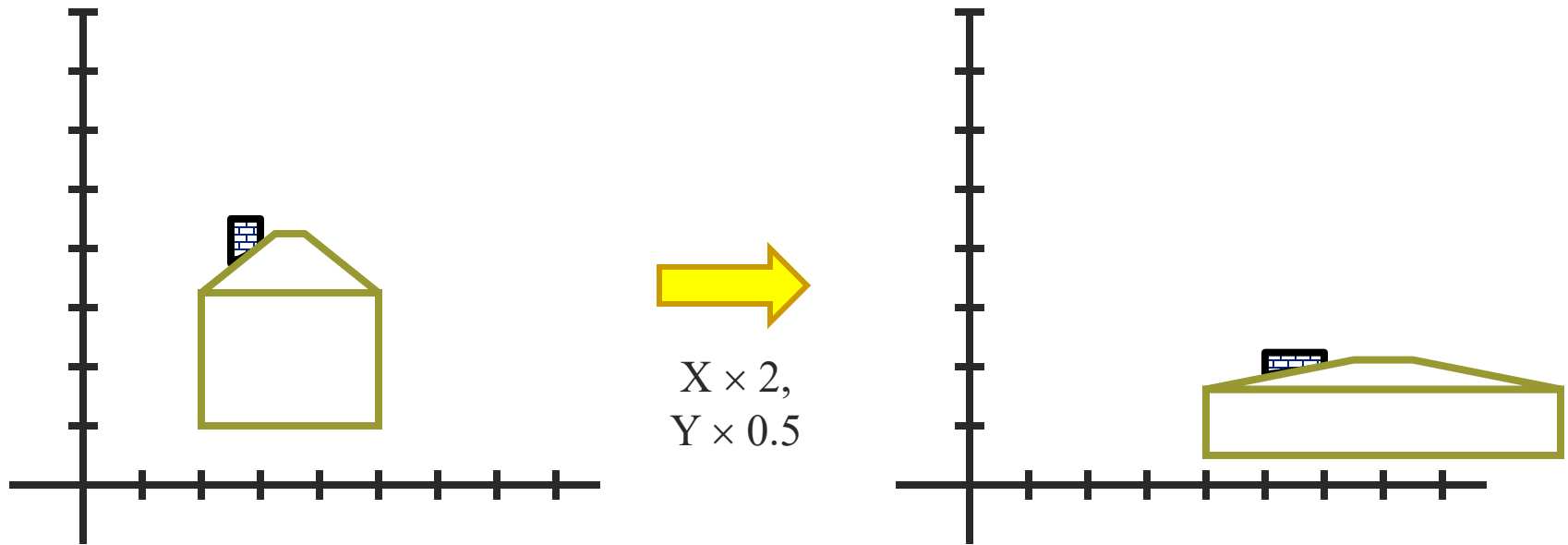




Scaling



- *Non-uniform scaling*: different scalars per component:





Scaling



- Scaling operation: $x' = ax$
 $y' = by$

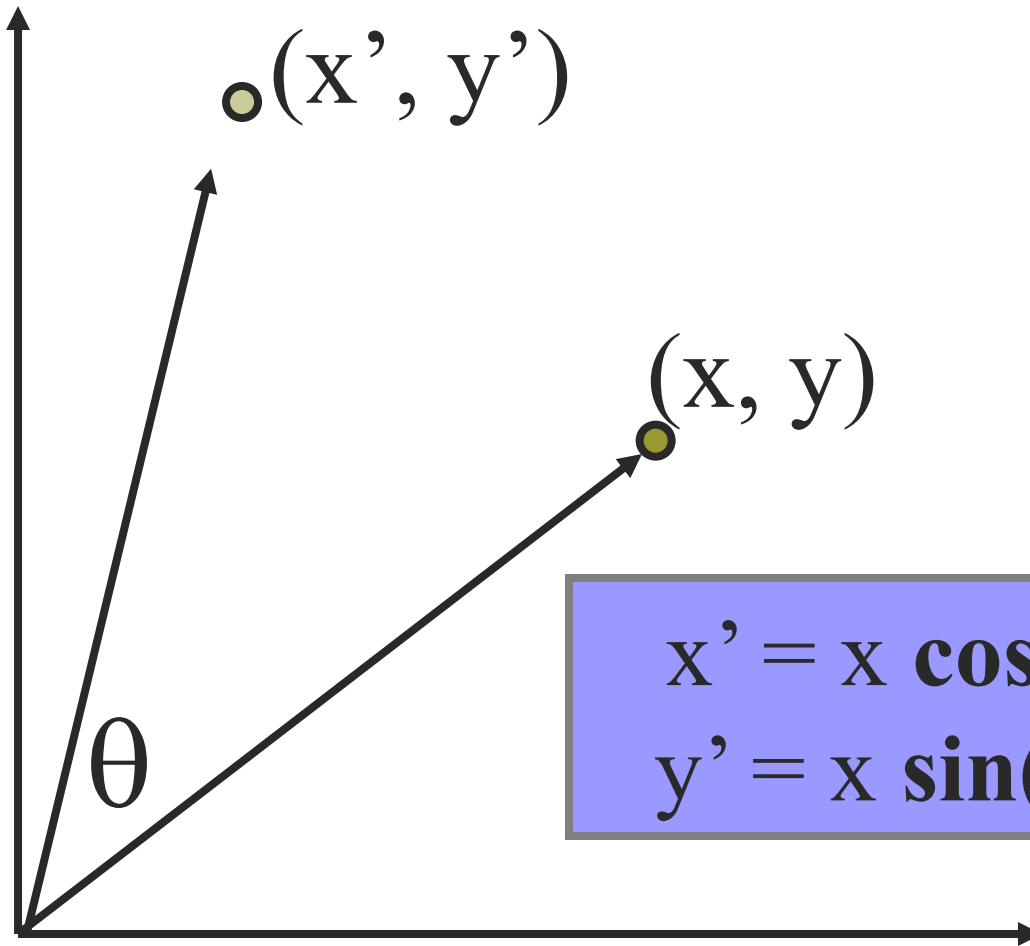
- Or, in matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}}_{\text{scaling matrix } S} \begin{bmatrix} x \\ y \end{bmatrix}$$

scaling matrix S



2-D Rotation



$$\begin{aligned}x' &= x \cos(\theta) - y \sin(\theta) \\y' &= x \sin(\theta) + y \cos(\theta)\end{aligned}$$



2-D Rotation



Polar coordinates...

$$x = r \cos(\phi)$$

$$y = r \sin(\phi)$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

Trig Identity...

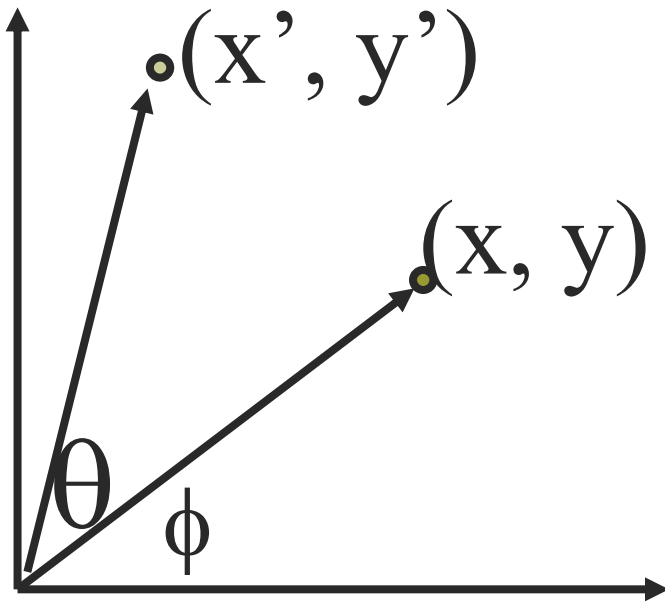
$$x' = r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta)$$

$$y' = r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta)$$

Substitute...

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$





2-D Rotation



This is easy to capture in matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} x \\ y \end{bmatrix}$$

Even though $\sin(\theta)$ and $\cos(\theta)$ are nonlinear functions of θ ,

- ***x' is a linear combination of x and y***
- ***y' is a linear combination of x and y***

What is the inverse transformation?

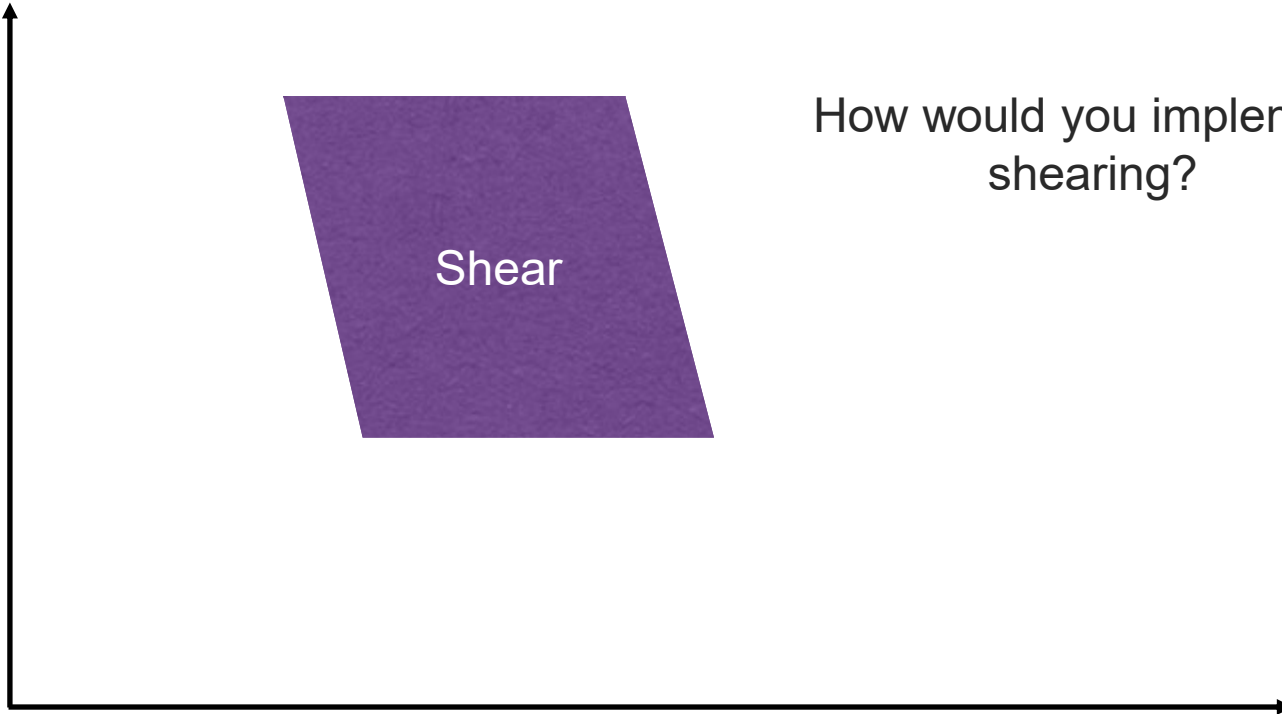
- Rotation by $-\theta$
- For rotation matrices $\mathbf{R}^{-1} = \mathbf{R}^T$



2-D Shearing



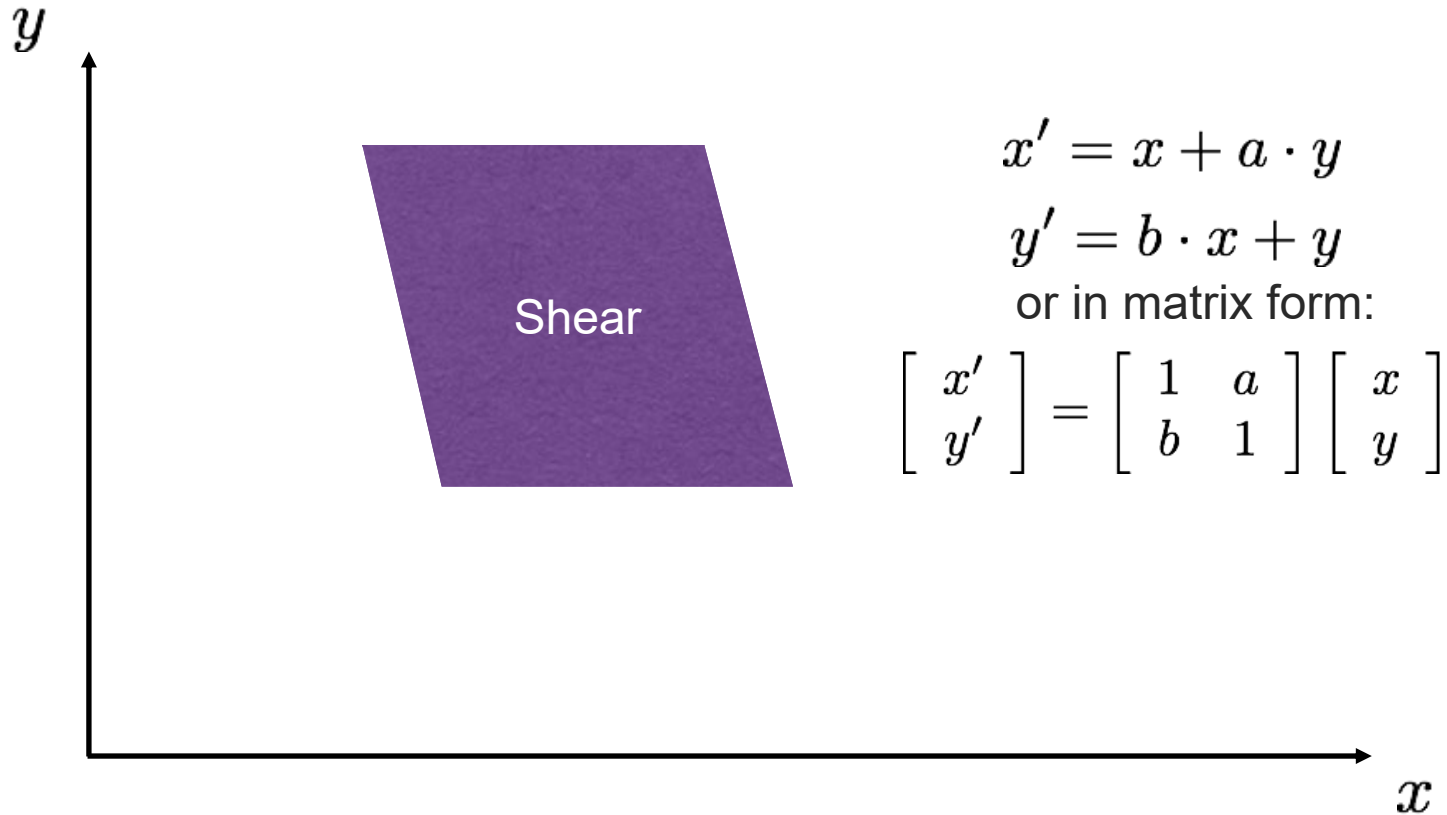
y



How would you implement shearing?



2-D Shearing





What transformations can be represented with a 2x2 matrix?



2D Scaling?

$$\mathbf{x}' = s_x * \mathbf{x}$$

$$\mathbf{y}' = s_y * \mathbf{y}$$

$$\begin{bmatrix} \mathbf{x}' \\ \mathbf{y}' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$$

2D Rotate around (0,0)?

$$\mathbf{x}' = \cos \Theta * \mathbf{x} - \sin \Theta * \mathbf{y}$$

$$\mathbf{y}' = \sin \Theta * \mathbf{x} + \cos \Theta * \mathbf{y}$$

$$\begin{bmatrix} \mathbf{x}' \\ \mathbf{y}' \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta \\ \sin \Theta & \cos \Theta \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$$

2D Shear?

$$\mathbf{x}' = \mathbf{x} + sh_x * \mathbf{y}$$

$$\mathbf{y}' = sh_y * \mathbf{x} + \mathbf{y}$$

$$\begin{bmatrix} \mathbf{x}' \\ \mathbf{y}' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$$



What transformations can be represented with a 2x2 matrix?



2D Mirror about Y axis?

$$\begin{aligned}x' &= -x \\ y' &= y\end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Mirror over (0,0)?

$$\begin{aligned}x' &= -x \\ y' &= -y\end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Translation?

$$\mathbf{x}' = \mathbf{x} + \mathbf{t}_x$$

$$\mathbf{y}' = \mathbf{y} + \mathbf{t}_y$$

NO!



2D Linear Transformations



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Only linear 2D transformations can be represented with a 2x2 matrix.
- Linear transformations are combinations of ...
 - Scale,
 - Rotation,
 - Shear, and
 - Mirror



Homogeneous coordinates



To convert to homogeneous coordinates:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$



Homogeneous Coordinates



- Q: How can we represent 2d translation as a 3x3 matrix using homogeneous coordinates?

$$\mathbf{x}' = \mathbf{x} + \mathbf{t}_x$$

$$\mathbf{y}' = \mathbf{y} + \mathbf{t}_y$$

- A: Using the rightmost column:

$$\mathbf{Translation} = \begin{bmatrix} 1 & 0 & \mathbf{t}_x \\ 0 & 1 & \mathbf{t}_y \\ 0 & 0 & 1 \end{bmatrix}$$



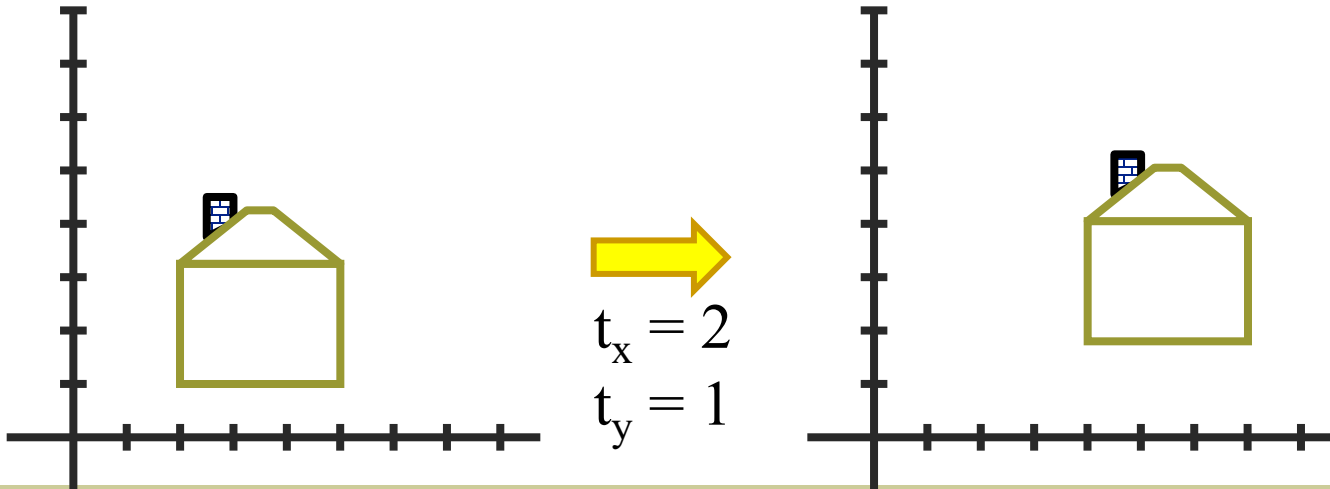
Translation



Homogeneous Coordinates



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$





Basic 2D Transformations



■ Basic 2D transformations as 3x3 matrices

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scale

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Shear

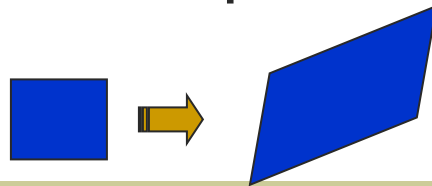


2D Affine Transformations



$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

- Affine transformations are combinations of ...
 - Linear transformations, and
 - Translations
- Parallel lines remain parallel





Affine Transformations



Affine transformations are combinations of

- Linear transformations, and
- Translations

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

or

Properties of affine transformations:

- Lines map to lines
- Parallel lines remain parallel
- Ratios are preserved
- Closed under composition

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Projective Transformations



Projective transformations are combos of

- Affine transformations, and
- Projective warps

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Properties of projective transformations:

- Lines map to lines
- Parallel lines do not necessarily remain parallel
- Ratios are not preserved
- Closed under composition
- Models change of basis
- Projective matrix is defined up to a scale (8 DOF)

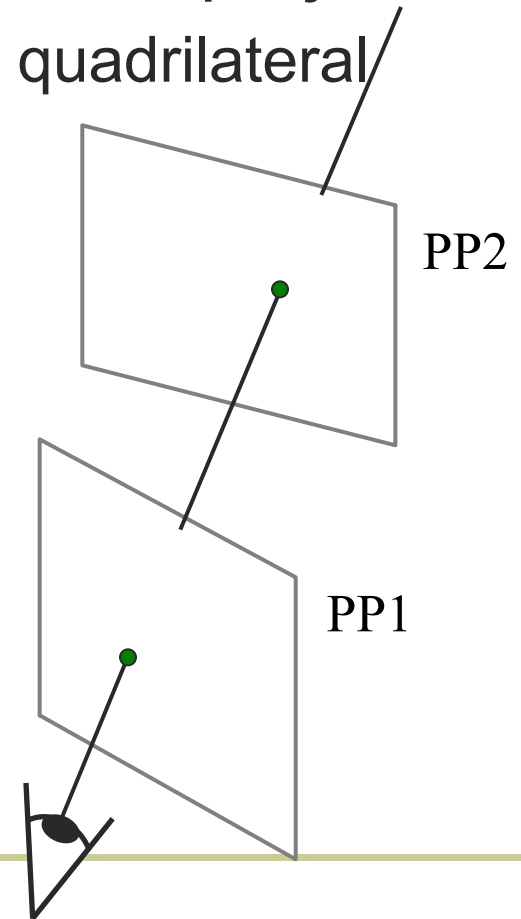


Image reprojection: Homography



- A projective transform is a mapping between any two PPs with the same center of projection
 - rectangle should map to arbitrary quadrilateral
 - parallel lines aren't
 - but must preserve straight lines
- called **Homography**

$$\begin{bmatrix} wx' \\ wy' \\ w \\ \mathbf{p}' \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ \mathbf{H} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \\ \mathbf{p} \end{bmatrix}$$





Homography

(x, y)



$$\left(\frac{wx'}{w}, \frac{wy'}{w} \right) = (x', y')$$

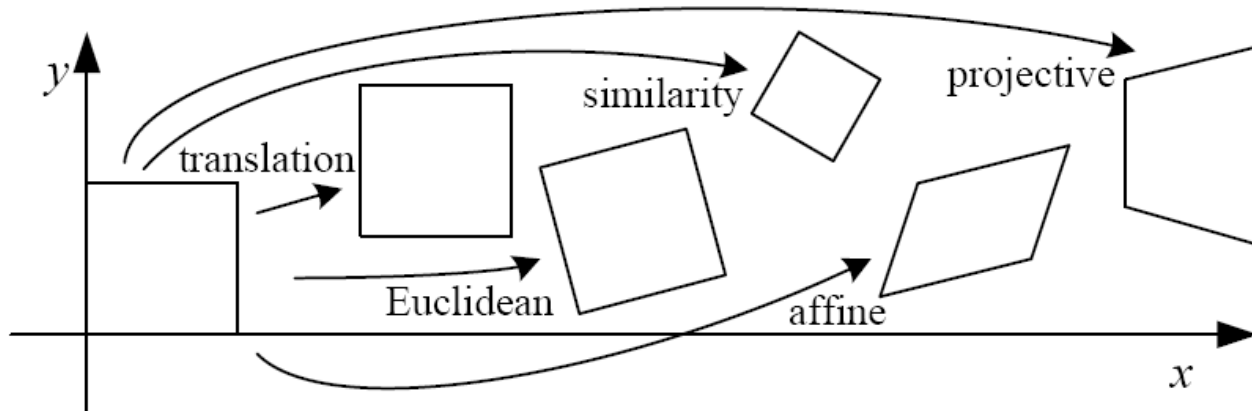
To **apply** a given homography **H**

- Compute $\mathbf{p}' = \mathbf{H}\mathbf{p}$ (regular matrix multiply)
- Convert $\mathbf{p}'$ from homogeneous to image coordinates

$$\begin{bmatrix} wx' \\ wy' \\ w \\ \mathbf{p}' \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ \mathbf{H} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \\ \mathbf{p} \end{bmatrix}$$



2D image transformations



Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} I & t \end{bmatrix}_{2 \times 3}$	2	orientation + ...	
rigid (Euclidean)	$\begin{bmatrix} R & t \end{bmatrix}_{2 \times 3}$	3	lengths + ...	
similarity	$\begin{bmatrix} sR & t \end{bmatrix}_{2 \times 3}$	4	angles + ...	
affine	$\begin{bmatrix} A \end{bmatrix}_{2 \times 3}$	6	parallelism + ...	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{3 \times 3}$	8	straight lines	



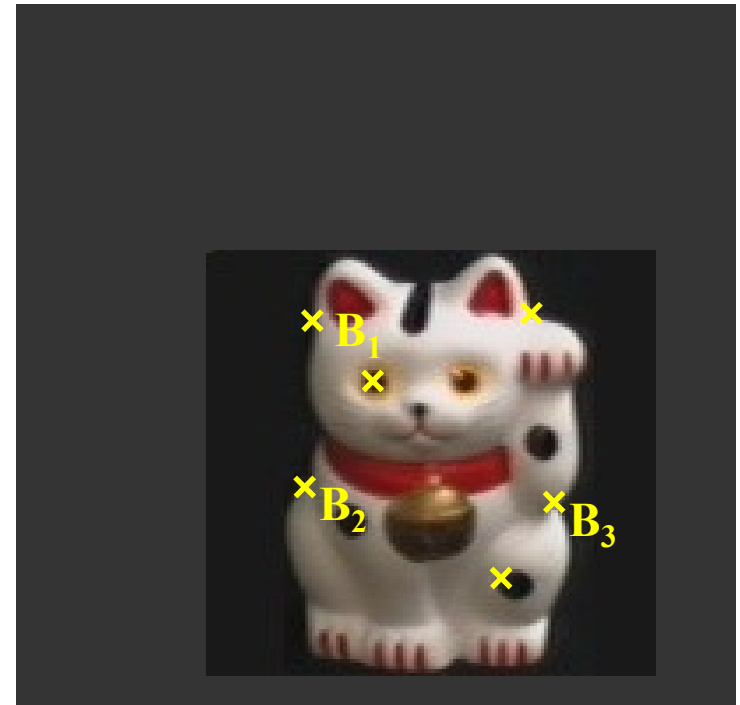
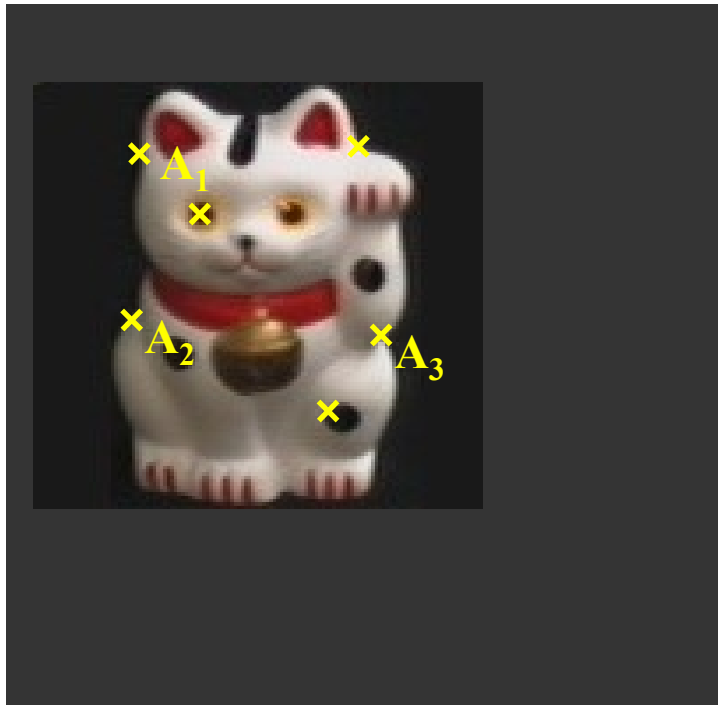
Today's class



- Introduction to alignment
- Alignment methods
 - Global methods
 - Hypothesize and test
- Image Transformation
 - Common image transformations
 - **Examples of solving image alignment**
- Homework: Mosaics



Example: solving for translation

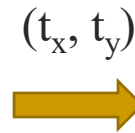
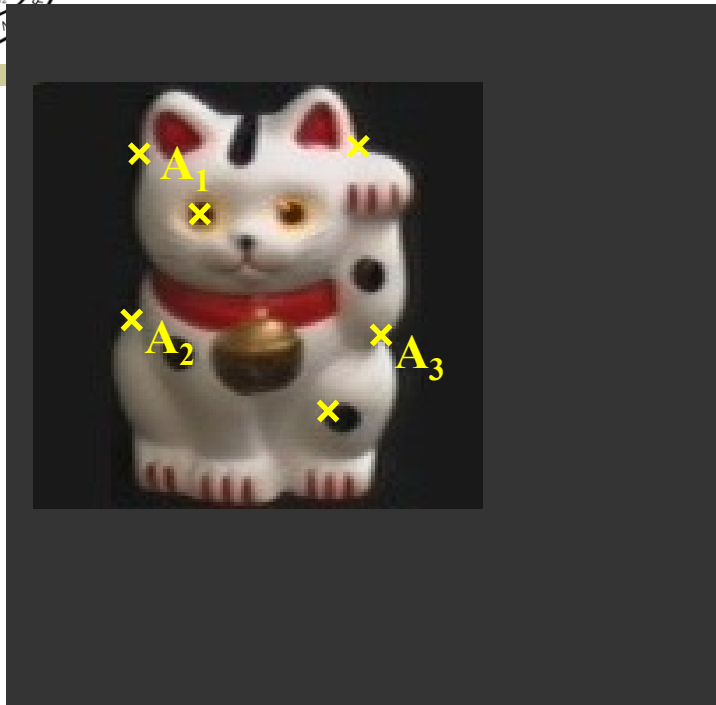


Given matched points in $\{A\}$ and $\{B\}$, estimate the translation of the object

$$\begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} = \begin{bmatrix} x_i^A \\ y_i^A \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$



Example: solving for translation



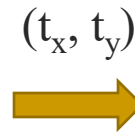
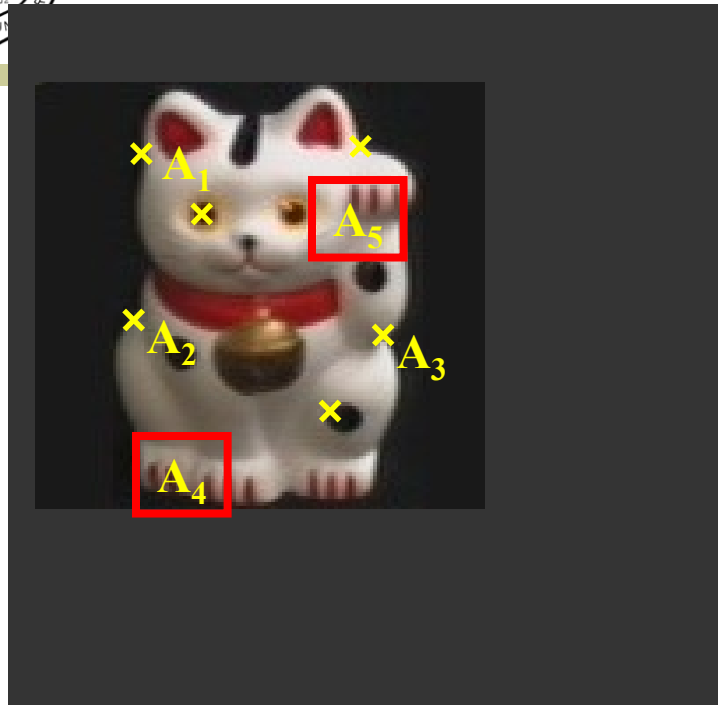
Least squares solution

1. Write down objective function
2. Derived solution
 - a) Compute derivative
 - b) Compute solution
3. Computational solution
 - a) Write in form $Ax=b$
 - b) Solve using pseudo-inverse or eigenvalue decomposition

$$\begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} = \begin{bmatrix} x_i^A \\ y_i^A \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \end{bmatrix} = \begin{bmatrix} x_1^B - x_1^A \\ y_1^B - y_1^A \\ \vdots \\ x_n^B - x_n^A \\ y_n^B - y_n^A \end{bmatrix}$$

Example: solving for translation



RANSAC solution

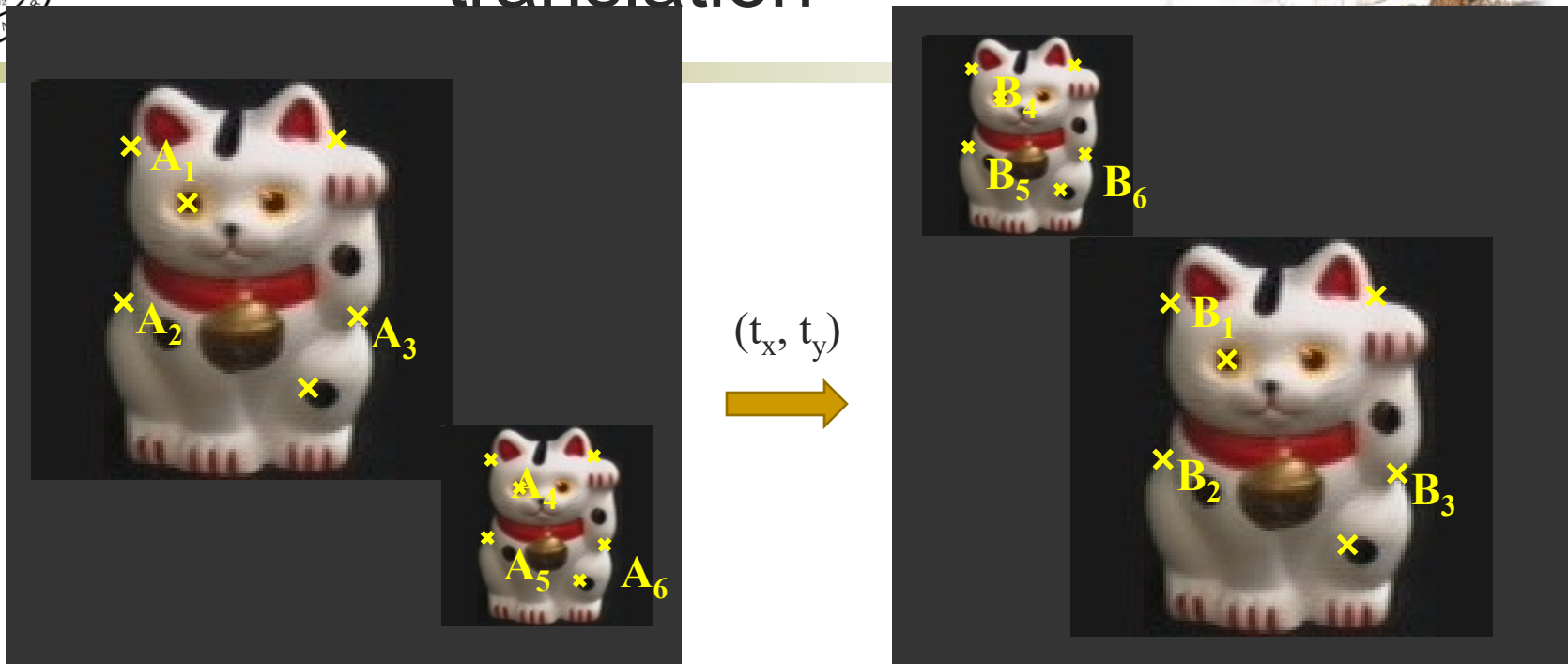
Problem: outliers

1. Sample a set of matching points (1 pair)
2. Solve for transformation parameters
3. Score parameters with number of inliers
4. Repeat steps 1-3 N times

$$\begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} = \begin{bmatrix} x_i^A \\ y_i^A \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$



Example: solving for translation



Problem: outliers, multiple objects, and/or many-to-one matches

Hough transform solution

1. Initialize a grid of parameter values
2. Each matched pair casts a vote for consistent values
3. Find the parameters with the most votes
4. Solve using least squares with inliers

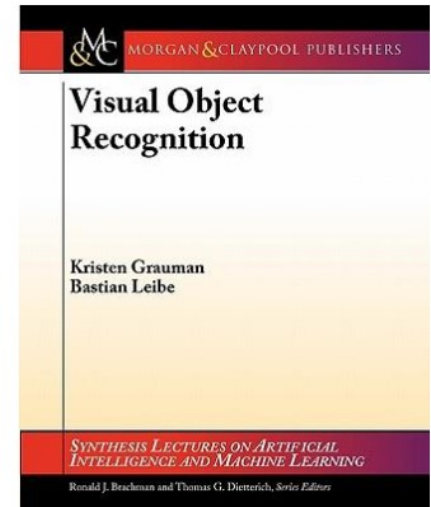
$$\begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} = \begin{bmatrix} x_i^A \\ y_i^A \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$



References



- We've created a script... for the part of the lecture on object recognition & categorization
 - K. Grauman, B. Leibe
Visual Object Recognition
Morgan & Claypool publishers, 2011
- Chapter 3: Local Feature Extraction (Last lecture)
- Chapter 5: Geometric Verification (Today)

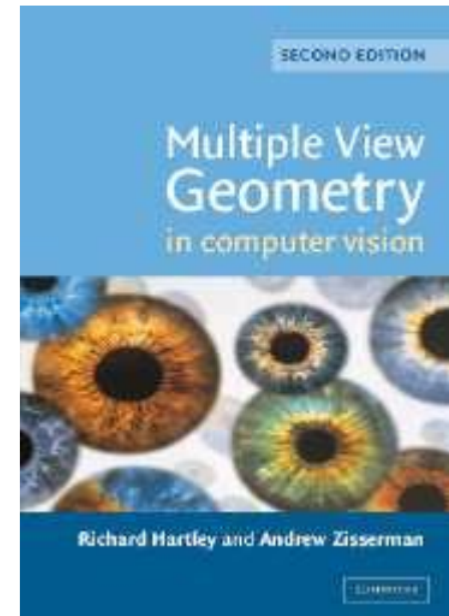




References



- More details on homography estimation can be found in Chapter 4.7 of
 - R. Hartley, A. Zisserman
Multiple View Geometry in Computer Vision
2nd Ed., Cambridge Univ. Press, 2004
- Details about the DoG detector and the SIFT descriptor can be found in
 - D. Lowe, [Distinctive image features from scale-invariant keypoints](#),
IJCV 60(2), pp. 91-110, 2004
- Try the available local feature detectors and descriptors
 - <http://www.robots.ox.ac.uk/~vgg/research/affine/detectors.html#binaries>





Today's class



- Introduction to alignment
- Alignment methods
 - Global methods
 - Hypothesize and test
- Image Transformation
 - Common image transformations
 - Examples of solving image alignment
- **Homework: Mosaics**



Homework: Mosaics



...

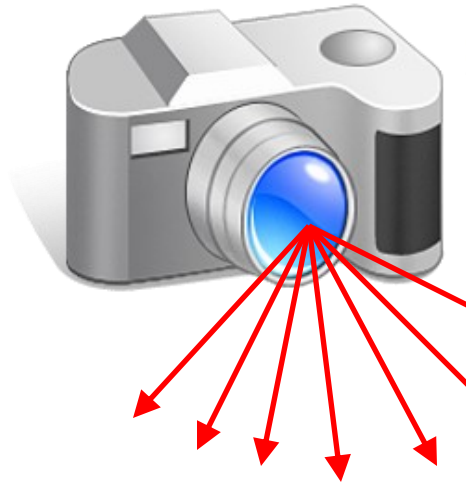


image from S. Seitz

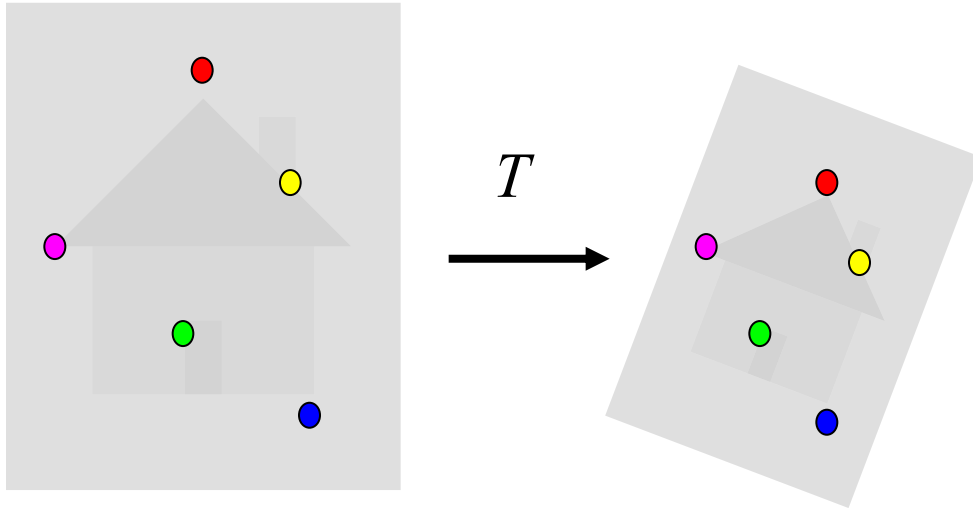
Obtain a wider angle view by combining multiple images.



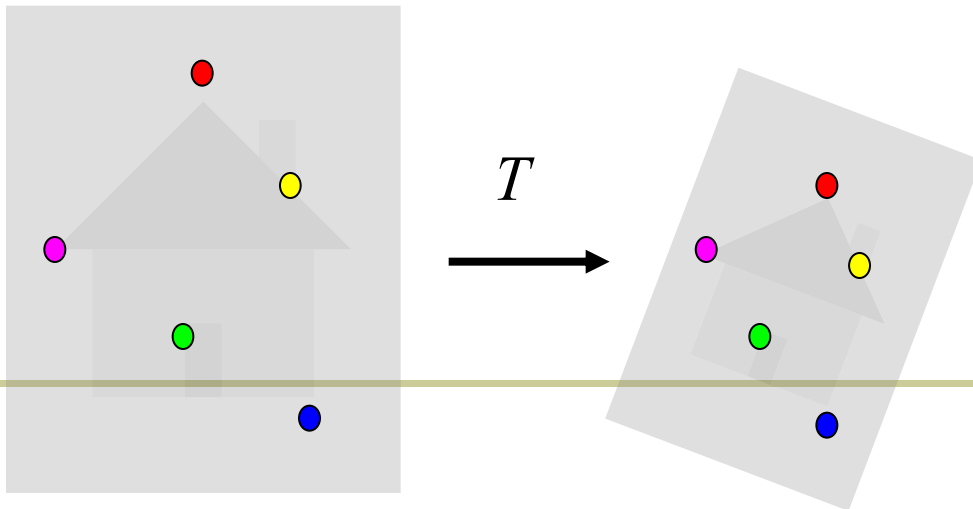
Main questions



Alignment: Given two images, what is the transformation between them?



Warping: Given a source image and a transformation, what does the transformed output look like?





How to stitch together a panorama (a.k.a. mosaic)?



■ Basic Procedure

- Take a sequence of images from the same position
 - Rotate the camera about its optical center
- Compute transformation (homography) between second image and first using corresponding points.
- Transform the second image to overlap with the first.
- Blend the two together to create a mosaic.
- (If there are more images, repeat)



本科生大作业



- 综述性研究报告：针对一个专门话题，进行深入调研分析，形成一篇高质量的综述。
- 培养：查阅文献，阅读文献，撰写文献的能力，可能启发对某个方向深入研究的兴趣。
- 提交格式：**CVPR**论文格式，建议英文撰写。
- 调研范围：顶级会议和顶级期刊论文，以及最新的**arXiv**论文，
- 不要仅仅去看一些公众号、知乎、中文博客
- 提交时间：**6月20号之前（毕业班）**



话题



- Low-level vision: denoise, super resolution etc.
- Edge detection
- Grouping and segmentation
- Local descriptor and image matching
- Deep learning for vision
- Tracking, video analysis
- Vision and language
- Weakly/Self supervised learning
- Etc.